

Geometry in the Complex Plane

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Introduction

The complex plane is similar to the Cartesian plane in the sense that each point on the plane is represented by a unique algebraic form. The main difference is that we only need one variable to represent a point in the complex plane instead of two. This sometimes makes calculations simpler, but some relationships among the points become less obvious.

Since using coordinates and complex numbers to solve geometry problems share a lot of common techniques, we shall mainly focus on the strength of complex numbers in this short note. We shall see that complex numbers work very well on problems involving circles and some simple transformations.

Section 1 includes the necessary knowledge in complex numbers, which we are supposed to know. Also, it covers the main geometric ingredients that we need to solve geometry problems. One should understand these things thoroughly before working on the problems.

Section 2 introduces the simple setting using the unit circle. Then in section 3, we shall see how the basic geometric transformations are interpreted using complex numbers.

1 Preliminary

In this section, we first provide the necessary knowledge about complex numbers and the complex plane. We shall introduce the relations between the complex plane and plane geometry.

1.1 Complex Numbers

The numbers we usually use are real numbers. But there is another kind of numbers called *complex numbers*(複數), which can be viewed as a generalization of the real number system. Consider the equation

$$x^2 + 1 = 0. \quad (1)$$

Since $x^2 \geq 0$ for any real number x , the equation has no real solution. We use an *imaginary number*(虛數) i to denote a solution of (1), so that $i^2 = -1$. Then we define a complex number to be a number of the form $a + bi$ where a and b are real numbers. The set of all complex numbers is denoted by $\mathbb{C}$.

Let $z = a + bi$ be a complex number. When $b = 0$, $z = a \in \mathbb{R}$ can be any real number. So each real number is in fact a complex number. In general, we call a the *real part*(實部) of z , denoted by $\text{Re}(z)$, and call b the *imaginary part*(虛部) of z , denoted by $\text{Im}(z)$. When $a = 0$, $z = bi$ is called a *purely imaginary number*(純虛數) (some may require $b \neq 0$). We use $i\mathbb{R}$ to denote the set of all purely imaginary numbers.

Two complex numbers are equal if and only if (iff) both their real parts and imaginary parts are the same. We can apply algebraic operations on complex numbers like what we do on real numbers. We may regard i as a variable and apply the fact $i^2 = -1$ whenever needed. Indeed, for any complex numbers $w = a + bi$ and $z = c + di$, we have

$$\begin{aligned} w + z &= (a + bi) + (c + di) = (a + c) + (b + d)i, \\ w - z &= (a + bi) - (c + di) = (a - c) + (b - d)i, \\ wz &= (a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i, \end{aligned}$$

and for $z \neq 0$ (i.e. $(c, d) \neq (0, 0)$),

$$\frac{w}{z} = \frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{ac - adi + bci - bdi^2}{c^2 - (di)^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i.$$

For every complex number $z = a + bi$, there are two important definitions associated to it. Firstly, the *complex conjugate*(共軛複數) of z is defined to be $a - bi$, and is denoted by $\bar{z}$. Secondly, the *modulus*(模) of z is defined to be $\sqrt{a^2 + b^2}$, and is denoted by $|z|$. Note that the modulus of a complex number must be a nonnegative real number. In particular, when z is a real number, then its modulus is simply its absolute value.

The following provides some basic facts about conjugates and moduli.

Proposition 1.1. Let z and w be complex numbers, and let a be a real number. Then the following hold.

- (a) $\overline{a} = a$
- (b) $\overline{\overline{z}} = z$
- (c) $\overline{az} = a\overline{z}$
- (d) $\operatorname{Re}(z) = \frac{z + \overline{z}}{2}$
- (e) $\operatorname{Im}(z) = \frac{z - \overline{z}}{2i}$
- (f) $\overline{z \pm w} = \overline{z} \pm \overline{w}$
- (g) $\overline{zw} = \overline{z} \times \overline{w}$
- (h) $\overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}}$
- (i) $|zw| = |z||w|$
- (j) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$
- (k) ([Triangle inequality](#)(三角不等式)) $|z + w| \leq |z| + |w|$
- (l) $|\overline{z}| = |z|$
- (m) $|z|^2 = z\overline{z}$

These facts help us handle conjugates and moduli. In short, conjugates behave well with the algebraic operations. Moduli also work well with multiplication. For addition, we can still use the triangle inequality to get a bound. Parts (d) and (e) provide us ways to represent the real part and the imaginary part of a complex number. As a corollary, we see that

$$z \in \mathbb{R} \quad \Leftrightarrow \quad \operatorname{Im}(z) = 0 \quad \Leftrightarrow \quad z = \overline{z}$$

and

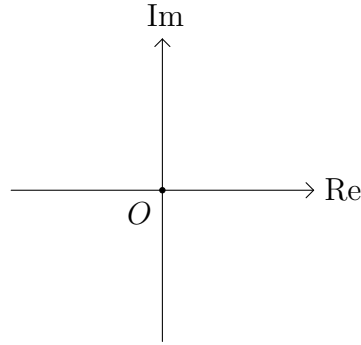
$$z \in i\mathbb{R} \quad \Leftrightarrow \quad \operatorname{Re}(z) = 0 \quad \Leftrightarrow \quad z + \overline{z} = 0.$$

Lastly, (m) is a useful result which connects the conjugates and moduli.

1.2 Complex Plane

The set of all real numbers is usually represented by a number line. Similarly, the set of all complex numbers can be regarded as a plane. The complex plane is similar to the Cartesian plane. Indeed, for each complex number $z = a + bi$, we assign its position to be the position of the point with coordinates (a, b) in the Cartesian plane. In other words, the x -coordinate and the y -coordinate are the real part and the imaginary part of z respectively. The horizontal x -axis is thus called the real axis, while the vertical y -axis is now called the imaginary axis. This is known as the *Argand diagram*(阿爾岡平面) or the *complex plane*(複數平面).

In this note, we always use small letters to denote the complex numbers of the corresponding points. For example, a, b, c are the complex numbers representing points A, B, C respectively.



In view of the definition, it is obvious that $Z(z)$ and $\overline{Z}(\bar{z})$ are mirror images of each other with respect to (w.r.t.) the real axis. Also, $|z| = \sqrt{a^2 + b^2}$ is the distance between $O(0)$ and Z .

Next, we have one more definition related to the geometric aspect of a complex number. For any nonzero complex number $z = a + bi$, we define its *argument*(輻角) to be the angle in radian measured from the positive real axis to the segment OZ in the anticlockwise direction, and is denoted by $\arg z$. We usually assume $0 \leq \arg z < 2\pi$. Note that we must have $\tan(\arg z) = \frac{b}{a}$ (except when $a = 0$, where $\tan \frac{\pi}{2}$ and $\tan \frac{3\pi}{2}$ are undefined). It can be verified that

$$\begin{aligned}\arg(zw) &\equiv \arg z + \arg w \pmod{2\pi}, \\ \arg\left(\frac{z}{w}\right) &\equiv \arg z - \arg w \pmod{2\pi}.\end{aligned}$$

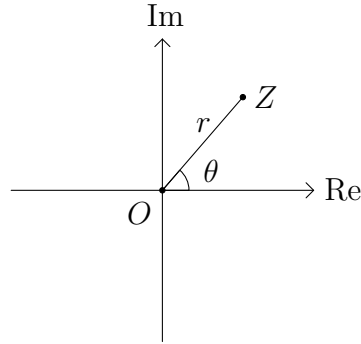
Note that $z \in \mathbb{R}$ iff $\arg z = 0$ or π , while $z \in i\mathbb{R}$ iff $\arg z = \frac{\pi}{2}$ or $\frac{3\pi}{2}$.

This gives rise to the polar form of a complex number (similar to the polar coordinates in the Cartesian plane). A complex number is usually written in the *standard form*(標準形式)

準式) $z = a + bi$. Its *polar form*(極式) is $z = r(\cos \theta + i \sin \theta)$ or simply $z = r \operatorname{cis} \theta$ where $r = |z|$ and $\theta = \arg z$. For $z = 0$, it is reasonable to define $r = 0$, while θ is arbitrary (which does not affect the value of $r \operatorname{cis} \theta$). For any nonzero complex numbers $z = r \operatorname{cis} \theta$ and $w = r' \operatorname{cis} \phi$ and any integer k , we have

$$\begin{aligned} zw &= rr' \operatorname{cis} (\theta + \phi), \\ \frac{z}{w} &= \frac{r}{r'} \operatorname{cis} (\theta - \phi), \\ z^k &= r^k \operatorname{cis} k\theta. \end{aligned}$$

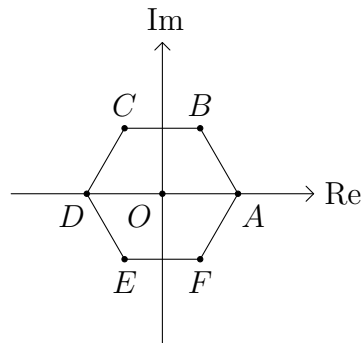
The last result comes from *De Moivre's theorem*(棣莫弗定理), which states that $(\operatorname{cis} \theta)^k = \operatorname{cis} k\theta$.



In addition, we can write a complex number in the *exponential form*(指數形式) $z = re^{i\theta}$ where r and θ are defined as above. Some algebraic operations can be easily carried out in this form using the law of indices. For example, for $z = re^{i\theta}$ and $w = r'e^{i\phi}$ and any integer k , we have

$$\begin{aligned} zw &= rr'e^{i(\theta+\phi)}, \\ \frac{z}{w} &= \frac{r}{r'}e^{i(\theta-\phi)}, \\ z^k &= r^ke^{i(k\theta)}. \end{aligned}$$

This is consistent to the results given in the polar form.



Next, the root of unity is another important concept. For any positive integer n , the n complex roots of the equation $z^n = 1$ are $z = \text{cis } \frac{2k\pi}{n}$ where $k = 0, 1, 2, \dots, n-1$. These are called the *n th roots of unity* (單位根). They are represented by the vertices of a regular n -gon with centre $O(0)$ in the complex plane. The modulus of every root of unity is 1.

From $z^n = 1$, we have $(z-1)(z^{n-1} + z^{n-2} + \dots + 1) = z^n - 1 = 0$. Therefore, for $z \neq 1$, we have

$$z^{n-1} + z^{n-2} + \dots + 1 = 0.$$

This is one of the simplest properties of the roots of unity.

1.3 Facts in the Complex Plane

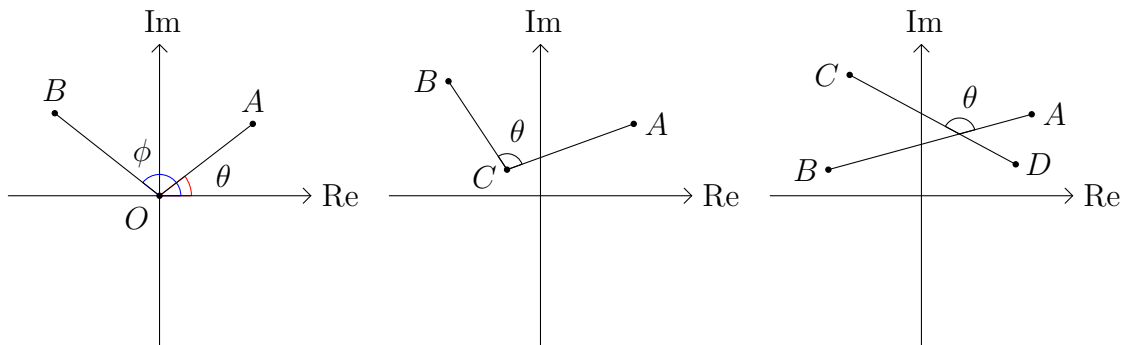
To use complex numbers to solve geometric problems, we first identify every point in the plane by a complex number using the complex plane. Then we try to work out the algebraic relations among these complex numbers to derive the corresponding geometric meaning of the results. For this purpose, we need to link the two things together. In this subsection, we shall introduce the basic facts we need to know.

1. (Distance) The distance between O and Z is $|z|$ as mentioned. One can generalize this to get $AB = |a - b|$ for any points $A(a)$ and $B(b)$.
2. (Angle) Let $\arg a = \theta$ and $\arg b = \phi$. Recall that

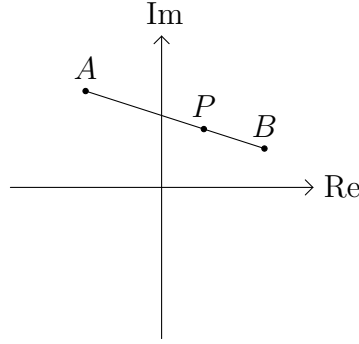
$$\arg \frac{b}{a} \equiv \arg b - \arg a = \phi - \theta \pmod{2\pi}.$$

Therefore, this measures the directed angle $\angle AOB$ (in the anticlockwise direction).

In general, $\arg \frac{b-c}{a-c}$ measures $\angle ACB$, while $\arg \frac{c-d}{a-b}$ measures the angle between the lines AB and CD , rotated from AB to CD .



3. (Sectional formula) Let P be a point on the segment AB with $AP : PB = r : s$. Then we have $p = \frac{rb + sa}{r + s}$. In particular, the midpoint of AB is represented by $\frac{a + b}{2}$. The formula is still valid for every point P on the line AB , where directed lengths are used to express the ratio.



4. (Straight line) We provide three forms of the equation of a straight line.

- (i) $\alpha z - \overline{\alpha z} = \beta i$ where $\alpha \in \mathbb{C}$ is nonzero and $\beta \in \mathbb{R}$

A point W lies on this straight line iff $\alpha w - \overline{\alpha w} = \beta i$. The corresponding equation in the Cartesian plane is $\text{Im}(\alpha)x + \text{Re}(\alpha)y = \frac{\beta}{2}$ where $z = x + yi$.

When solving equations of two straight lines, one may regard z and $\bar{z}$ as different variables and solve for z . Two lines with equations $\alpha z - \overline{\alpha z} = \beta i$ and $\alpha' z - \overline{\alpha' z} = \beta' i$ are parallel iff $\frac{\alpha'}{\alpha} \in \mathbb{R}$.

- (ii) $\frac{z - a}{b - a} = \frac{\bar{z} - \bar{a}}{\bar{b} - \bar{a}}$ where $a, b \in \mathbb{C}$ are distinct

This is the equation of the line joining two distinct points A and B . To verify this, we see that

$$\begin{aligned}
 & A, B, Z \text{ are collinear} \\
 \Leftrightarrow & \angle BAZ = 0 \text{ or } \pi \\
 \Leftrightarrow & \frac{z - a}{b - a} \in \mathbb{R} \\
 \Leftrightarrow & \frac{z - a}{b - a} = \overline{\left(\frac{z - a}{b - a} \right)} = \frac{\bar{z} - \bar{a}}{\bar{b} - \bar{a}}.
 \end{aligned}$$

This can be converted to the first form via expansion, which gives

$$(\bar{b} - \bar{a})z - (b - a)\bar{z} = a\bar{b} - \bar{a}b.$$

Here, $a\bar{b} - \bar{a}b = a\bar{b} - \overline{a\bar{b}} = 2i\text{Im}(a\bar{b})$ must be purely imaginary.

(iii) $ta + (1 - t)b$ where $t \in \mathbb{R}$ and $a, b \in \mathbb{C}$ are distinct

This is the locus of points lying on the line AB by the sectional formula. We seldom use this form to represent a straight line as it is difficult to solve equations with the constraint that t is real. However, this is useful in case we need to represent a variable point lying on AB .

5. (Circle) The equation of a circle with centre A and radius r is $|z - a| = r$ by the distance formula. Note that we can remove the modulus as follows.

$$\begin{aligned} & |z - a| = r \\ \Leftrightarrow & |z - a|^2 = r^2 \\ \Leftrightarrow & (z - a)\overline{(z - a)} = r^2 \\ \Leftrightarrow & z\bar{z} - \bar{a}z - a\bar{z} + (a\bar{a} - r^2) = 0 \end{aligned}$$

6. (Concyclic points) Given four distinct points A, B, C, D , then A, B, C, D are concyclic iff $\angle ABC = \angle ADC$ (using directed angles). Now, we have

$$\begin{aligned} & \angle ABC = \angle ADC \\ \Leftrightarrow & \arg \frac{c - b}{a - b} = \arg \frac{c - d}{a - d} \\ \Leftrightarrow & \arg \left(\frac{c - b}{a - b} \div \frac{c - d}{a - d} \right) = 0 \\ \Leftrightarrow & \frac{c - b}{a - b} \div \frac{c - d}{a - d} \in \mathbb{R}. \end{aligned}$$

Note that this includes the limiting case when A, B, C, D are collinear (thus lying on an infinite circle), which can usually be rejected easily.

7. (Area) The area of a convex polygon $Z_1 Z_2 \cdots Z_n$ in anticlockwise orientation is

$$\frac{1}{2} \text{Im}(\bar{z}_1 z_2 + \bar{z}_2 z_3 + \cdots + \bar{z}_n z_1).$$

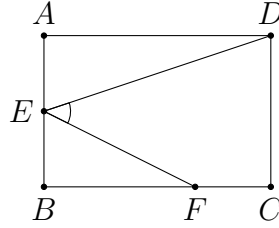
This can be deduced via the shoelace formula in coordinate geometry.

We first try to use the basic concepts to illustrate how to use complex numbers to solve geometric problems.

Problem 1.1. Let $ABCD$ be a rectangle with $AB = 2$ and $BC = 3$. Let E be the midpoint of AB and let F be a point on BC satisfying $BF = 2$. Find $\angle FED$.

Analysis. This problem demonstrates the use of complex plane to evaluate angles.

Solution.



Put the vertices on the complex plane such that $b = 0$, $c = 3$, $a = 2i$ and $d = 3 + 2i$. Obviously, $e = \frac{a+b}{2} = i$ and $f = 2$. In view of the orientation,

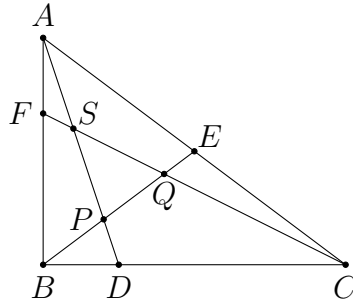
$$\angle FED = \arg \frac{d-e}{f-e} = \arg \frac{(3+2i)-i}{2-i} = \arg \frac{3+i}{2-i} = \arg (1+i) = \frac{\pi}{4}.$$

The last equality is true since $\tan^{-1} \frac{1}{1} = \frac{\pi}{4}$ and $1+i$ lies in the first quadrant. In other words, the angle is 45° . $\square$

Problem 1.2. In $\triangle ABC$, $AB = 6$, $BC = 8$ and $CA = 10$. The points D, E, F lie on the sides BC, CA, AB respectively such that $BD : DC = 1 : 3$, $CE = EA$ and $AF : FB = 1 : 2$. Suppose AD meets BE at P , BE meets CF at Q , CF meets AD at S . Find the area of $\triangle PQS$.

Analysis. Using this simple example, we demonstrate how to find the equations of straight lines and the intersection points of the lines. Then we apply the formula to find the area.

Solution.



Note that $\angle ABC = 90^\circ$ by Pythagoras' theorem. So we can put the vertices on the complex plane so that $b = 0$, $c = 8$ and $a = 6i$. Thus, we have

$$\begin{aligned} d &= \frac{(1)(8) + (3)(0)}{1+3} = 2, \\ e &= \frac{8+6i}{2} = 4+3i, \\ f &= \frac{(1)(0) + (2)(6i)}{1+2} = 4i. \end{aligned}$$

Next, we find the equations of AD, BE, CF as follows.

$$\begin{aligned} AD : \quad & \frac{z-2}{6i-2} = \frac{\bar{z}-\bar{2}}{\bar{6i}-\bar{2}} = \frac{\bar{z}-2}{-6i-2} \\ \Leftrightarrow \quad & (1+3i)z - (1-3i)\bar{z} = 12i, \end{aligned} \tag{1}$$

$$\begin{aligned} BE : \quad & \frac{z-0}{(4+3i)-0} = \frac{\bar{z}-0}{(4-3i)-0} \\ \Leftrightarrow \quad & (4-3i)z - (4+3i)\bar{z} = 0, \end{aligned} \tag{2}$$

$$\begin{aligned} CF : \quad & \frac{z-8}{4i-8} = \frac{\bar{z}-8}{-4i-8} \\ \Leftrightarrow \quad & (2+i)z - (2-i)\bar{z} = 16i. \end{aligned}$$

We solve the equations (1) and (2). This means we combine the equations to eliminate the term $\bar{z}$. This gives an expression for z alone. Since the intersection of AD and BE is P , the expression is the complex number p . We get

$$p = \frac{(4+3i)(12i) - (1-3i)(0)}{(4+3i)(1+3i) - (1-3i)(4-3i)} = \frac{-36+48i}{30i} = \frac{8+6i}{5}.$$

Similarly,

$$\begin{aligned} q &= \frac{(2-i)(0) - (4+3i)(16i)}{(2-i)(4-3i) - (4+3i)(2+i)} = \frac{48-64i}{-20i} = \frac{16+12i}{5}, \\ s &= \frac{(2-i)(12i) - (1-3i)(16i)}{(2-i)(1+3i) - (1-3i)(2+i)} = \frac{-36+8i}{10i} = \frac{4+18i}{5}. \end{aligned}$$

We can now compute the area of $\triangle PQS$, which is

$$\begin{aligned} & \frac{1}{2} \text{Im}(\bar{p}q + \bar{q}s + \bar{s}p) \\ &= \frac{1}{2} \text{Im}\left(\frac{8-6i}{5} \cdot \frac{16+12i}{5} + \frac{16-12i}{5} \cdot \frac{4+18i}{5} + \frac{4-18i}{5} \cdot \frac{8+6i}{5}\right) \\ &= \frac{1}{2} \text{Im}\left(\frac{200}{25} + \frac{280+240i}{25} + \frac{140-120i}{25}\right) \\ &= \frac{12}{5}. \end{aligned}$$

□

Problem 1.3. (Apollonius circle) Let A and B be two distinct points and let $x : y$ be a fixed ratio where x and y are distinct positive real numbers. Prove that the locus of points P satisfying $PA : PB = x : y$ is a circle.

Analysis. The circle is called the Apollonius circle. Using complex numbers, it suffices to show the equation of the locus has the form $|z - w| = r$.

Solution. It is given that $|p - a| : |p - b| = x : y$. Having an equation with moduli is not simple. We usually square both sides to convert them to conjugates by [proposition 1.1\(m\)](#). Note that

$$\begin{aligned} & \left| \frac{p - a}{p - b} \right| = \frac{x}{y} \\ \Leftrightarrow & y^2 |p - a|^2 = x^2 |p - b|^2 \\ \Leftrightarrow & y^2 (p - a)(\bar{p} - \bar{a}) = x^2 (p - b)(\bar{p} - \bar{b}) \\ \Leftrightarrow & (y^2 - x^2)p\bar{p} + (x^2\bar{b} - y^2\bar{a})p + (x^2b - y^2a)\bar{p} + y^2a\bar{a} - x^2b\bar{b} = 0. \end{aligned}$$

Recall that the equation of a circle is $z\bar{z} - \bar{w}z - w\bar{z} + (w\bar{w} - r^2) = 0$, which comes from the expansion of $|z - w| = r$. Here, the variable z should correspond to p , as P is the variable point.

By comparing like terms, we would like to have $w = \frac{y^2a - x^2b}{y^2 - x^2}$ and

$$\begin{aligned} r^2 &= \left(\frac{y^2a - x^2b}{y^2 - x^2} \right) \left(\frac{y^2\bar{a} - x^2\bar{b}}{y^2 - x^2} \right) - \frac{y^2a\bar{a} - x^2b\bar{b}}{y^2 - x^2} \\ &= \frac{\textcolor{red}{y}^4a\bar{a} - x^2y^2(\bar{a}b + a\bar{b}) + \textcolor{blue}{x}^4b\bar{b}}{(y^2 - x^2)^2} - \frac{\textcolor{red}{y}^4a\bar{a} - x^2y^2(a\bar{a} + b\bar{b}) + \textcolor{blue}{x}^4b\bar{b}}{(y^2 - x^2)^2} \\ &= \frac{x^2y^2(a\bar{a} - \bar{a}b - a\bar{b} + b\bar{b})}{(y^2 - x^2)^2} \\ &= \frac{x^2y^2(a - b)(\bar{a} - \bar{b})}{(y^2 - x^2)^2} \\ &= \frac{x^2y^2|a - b|^2}{(y^2 - x^2)^2}. \end{aligned}$$

Since $r > 0$, we should take $r = \frac{xy|a - b|}{|y^2 - x^2|}$, which is clearly a positive real number.

Thus, the locus becomes

$$\left| p - \frac{y^2a - x^2b}{y^2 - x^2} \right| = \frac{xy|a - b|}{|y^2 - x^2|},$$

which is the equation of a circle with centre $\frac{y^2a - x^2b}{y^2 - x^2}$ and radius $\frac{xy|a - b|}{|y^2 - x^2|}$. □

Remarks. Note that the centre of the Apollonius circle lies on the line AB (one may see this easily as the equation of this line is $sa + tb$ with $s + t = 1$).

If $x = y$, the locus is the perpendicular bisector of AB . If one of x, y is 0, then the locus is the single point A or B . All these cases can be regarded as degenerate Apollonius circles.

Problem 1.4. (Ptolemy's theorem) Let $ABCD$ be a convex quadrilateral. Prove that

$$AB \times CD + BC \times AD \geq AC \times BD,$$

and equality holds when A, B, C, D are concyclic.

Analysis. We demonstrate the use of the triangle inequality to prove this geometric inequality.

Solution. Using complex numbers, we need to prove

$$|a - b| |c - d| + |b - c| |a - d| \geq |a - c| |b - d|.$$

Indeed, by the triangle inequality, we have

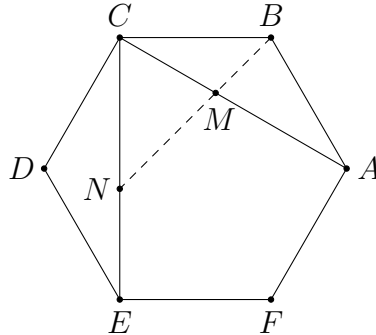
$$\begin{aligned} |a - b| |c - d| + |b - c| |a - d| &= |ac - bc - ad + bd| + |ab - ac - bd + cd| \\ &\geq |(ac - bc - ad + bd) + (ab - ac - bd + cd)| \\ &= |ab - bc - ad + cd| \\ &= |a - c| |b - d|. \end{aligned}$$

Equality holds when $\arg(a - b)(c - d) = \arg(b - c)(a - d)$ (since in that case the three points involved in the triangle inequality form a degenerate triangle), i.e. $\frac{(a - b)(c - d)}{(b - c)(a - d)} \in \mathbb{R}^+$. This is equivalent to A, B, C, D being concyclic. $\square$

Problem 1.5. (IMO Prelim 1998 Problem 6) $ABCDEF$ is a regular hexagon. M, N are points on the segments AC, CE respectively such that $\frac{AM}{AC} = \frac{CN}{CE} = r$. If B, M, N are collinear, find r .

Analysis. The vertices of a regular polygon can be easily expressed in terms of roots of unity.

Solution.



Let $\omega = \text{cis } \frac{2\pi}{6}$ be the root of unity such that $\omega^6 = 1$. Then we define $a = 1$, $b = \omega$, $c = \omega^2$, $d = \omega^3 = -1$, $e = \omega^4$ and $f = \omega^5$. From $\frac{AM}{AC} = r$, we have $\frac{AM}{MC} = \frac{r}{1-r}$. This gives

$$m = r\omega^2 + (1-r),$$

and similarly

$$n = r\omega^4 + (1-r)\omega^2.$$

Note that $\bar{\omega} = \frac{|\omega|^2}{\omega} = \frac{1}{\omega}$ (so this is a common property of all roots of unity). Thus, we find the equation of BM as follows.

$$\begin{aligned} & \frac{z - \omega}{r\omega^2 + (1-r) - \omega} = \frac{\bar{z} - \bar{\omega}}{r\bar{\omega}^2 + (1-r) - \bar{\omega}} \\ \Leftrightarrow & \frac{z - \omega}{r\omega^2 + (1-r) - \omega} = \frac{\omega^2 \bar{z} - \omega}{r + (1-r)\omega^2 - \omega} \\ \Leftrightarrow & \frac{z - \omega}{(1-\omega)(1-r-r\omega)} = \frac{\omega^2 \bar{z} - \omega}{(1-\omega)(r - (1-r)\omega)} \\ \Leftrightarrow & (r - (1-r)\omega)z - (1-r-r\omega)\omega^2 \bar{z} = -(1-2r)\omega - (1-2r)\omega^2 \end{aligned}$$

Since N lies on BM , we can put $z = n$. Noting $\omega^3 = -1$, we have

$$\begin{aligned} & (r - (1-r)\omega)((1-r)\omega^2 + r\omega^4) \\ & - (1-r-r\omega)\omega^2((1-r)\bar{\omega}^2 + r\bar{\omega}^4) = -(1-2r)\omega - (1-2r)\omega^2 \\ \Rightarrow & r(1-r)\omega^4 - (1-r)^2\omega^5 + r^2\omega^6 - r(1-r)\omega^7 \\ & - r(1-r) + r^2\omega - (1-r)^2\omega^2 + r(1-r)\omega^3 = -(1-2r)\omega^3 - (1-2r)\omega^4 \\ \Rightarrow & (-1 + 3r^2) + (-1 + 3r^2)\omega = 0 \\ \Rightarrow & (3r^2 - 1)(1 + \omega) = 0. \end{aligned}$$

Since $1 + \omega \neq 0$, we must have $3r^2 - 1 = 0$. As $r > 0$, the only solution is $r = \frac{1}{\sqrt{3}}$. $\square$

2 Unit Circle

One advantage of using complex numbers over coordinate geometry is to solve geometric problems with a circle. Coordinate geometry is not suitable to handle circles because radicals are usually involved. However, in complex numbers, we shall see a convenient way to handle this kind of problems.

If the figure only consists of one circle, or one of the circles plays the main role, then we can take this circle to be the unit circle in the complex plane. This means we set it as the circle with centre 0 and radius 1. In this setting, many things will become much simpler.

1. (Point on the unit circle) Any point Z on the unit circle satisfies $|z| = 1$, as this is simply the equation of the unit circle. This is equivalent to $z\bar{z} = |z|^2 = 1$, which gives a simple and useful relationship between z and $\bar{z}$.
2. (Chord of the unit circle) For any points A, B on the unit circle, the equation of the line AB is

$$z + ab\bar{z} = a + b.$$

This can be derived from the formula $\frac{z-a}{b-a} = \frac{\bar{z}-\bar{a}}{\bar{b}-\bar{a}}$ together with the facts $\bar{a} = \frac{1}{a}$ and $\bar{b} = \frac{1}{b}$.

3. (Tangent to the unit circle) Taking the limiting case of the chord when B tends to A , we get the equation of the tangent at A to the unit circle, which is

$$z + a^2\bar{z} = 2a.$$

4. (Perpendicular line to a chord) For any points A, B on the unit circle and any other point C in the complex plane, the equation of the line passing through C and perpendicular to AB is

$$z - ab\bar{z} = c - ab\bar{c}.$$

To see this, it suffices to consider the locus of points Z such that $\arg\left(\frac{z-c}{b-a}\right) = \frac{\pi}{2}$ or $\frac{3\pi}{2}$. This holds iff $\frac{z-c}{b-a} \in i\mathbb{R}$ iff

$$\frac{z-c}{b-a} + \frac{\bar{z}-\bar{c}}{\bar{b}-\bar{a}} = 0.$$

Again, by using $\bar{a} = \frac{1}{a}$ and $\bar{b} = \frac{1}{b}$, we get the desired formula after simplification.

Note that the limiting case $B = A$ also works, which gives the equation

$$z - a^2\bar{z} = c - a^2\bar{c}.$$

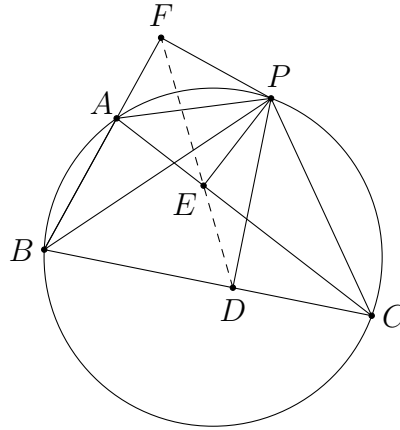
5. (Triangle geometry) Suppose the circumcircle of $\triangle ABC$ is the unit circle. Then its circumcentre is 0, its centroid is $\frac{a+b+c}{3}$ (which is true for any $\triangle ABC$) and its orthocentre is $a+b+c$ by the property of the Euler line.

We now consider some applications of complex numbers in solving geometric problems with circles.

Problem 2.1. (Simson line) Let P be a point on the circumcircle of $\triangle ABC$. Let D, E, F be the feet of perpendiculars from P to the sidelines BC, CA, AB respectively. Prove that D, E, F lie on the same line ℓ . Also, let H be the orthocentre of $\triangle ABC$. Prove that ℓ bisects PH .

Analysis. The line ℓ is called the Simson line of P w.r.t. $\triangle ABC$. We shall take the circumcircle to be the unit circle and demonstrate how to prove this kind of statements.

Solution.



Suppose A, B, C, P lie on the unit circle. We have

$$BC : z + bc\bar{z} = b + c, \quad (1)$$

$$PD : z - bc\bar{z} = p - bc\bar{p}. \quad (2)$$

Solving (1) and (2), we get the complex number corresponding to D . Indeed, we have

$$d = \frac{b + c + p - bc\bar{p}}{2}.$$

Note that we do not need to solve for E and F again, since we can get immediately by symmetry that

$$e = \frac{c + a + p - ca\bar{p}}{2}, \quad f = \frac{a + b + p - ab\bar{p}}{2}.$$

Now, we find the equation of the line DE , which is $\frac{z - d}{e - d} = \frac{\bar{z} - \bar{d}}{\bar{e} - \bar{d}}$. Note that

$$\begin{aligned} \frac{z - d}{e - d} &= \frac{\bar{z} - \bar{d}}{\bar{e} - \bar{d}} \\ \Leftrightarrow \frac{z - \frac{b + c + p - bc\bar{p}}{2}}{\frac{c + a + p - ca\bar{p}}{2} - \frac{b + c + p - bc\bar{p}}{2}} &= \frac{\bar{z} - \left(\frac{b + c + p - bc\bar{p}}{2}\right)}{\left(\frac{c + a + p - ca\bar{p}}{2}\right) - \left(\frac{b + c + p - bc\bar{p}}{2}\right)} \\ \Leftrightarrow \frac{2z - (b + c + p - bc\bar{p})}{a - b + c(b - a)\bar{p}} &= \frac{2\bar{z} - (\bar{b} + \bar{c} + \bar{p} - \bar{b}c\bar{p})}{\bar{a} - \bar{b} + \bar{c}(\bar{b} - \bar{a})p}. \end{aligned}$$

(Then we use the relations $\bar{a} = \frac{1}{a}$, $\bar{b} = \frac{1}{b}$, $\bar{c} = \frac{1}{c}$ and $\bar{p} = \frac{1}{p}$.)

$$\begin{aligned} \Leftrightarrow \frac{2z - \left(b + c + p - bc\frac{1}{p}\right)}{a - b + c(b - a)\frac{1}{p}} &= \frac{2\bar{z} - \left(\frac{1}{b} + \frac{1}{c} + \frac{1}{p} - \frac{1}{bc}p\right)}{\frac{1}{a} - \frac{1}{b} + \frac{1}{c}\left(\frac{1}{b} - \frac{1}{a}\right)p} \\ \Leftrightarrow \frac{2pz - (pb + pc + p^2 - bc)}{p(a - b) + c(b - a)} &= \frac{2abcp\bar{z} - a(cp + bp + bc - p^2)}{cp(b - a) + (a - b)p^2} \\ \Leftrightarrow \frac{2pz - (pb + pc + p^2 - bc)}{(a - b)(p - c)} &= \frac{2abcp\bar{z} - a(cp + bp + bc - p^2)}{p(a - b)(p - c)} \\ \Leftrightarrow 2p^2z - 2abcp\bar{z} &= p(pb + pc + p^2 - bc) - a(cp + bp + bc - p^2) \\ \Leftrightarrow 2p^2z - 2abcp\bar{z} &= p^3 + (a + b + c)p^2 - (ab + bc + ca)p - abc. \end{aligned} \quad (3)$$

As this equation is symmetric in a, b, c , the lines EF and FD also have the same equation. This means D, E, F are collinear.

Next, the orthocentre H satisfies $h = a + b + c$ as explained. Therefore, the midpoint of PH , which we denote by Q , satisfies $q = \frac{a + b + c + p}{2}$. It suffices to show q satisfies the equation (3). Indeed, we check that

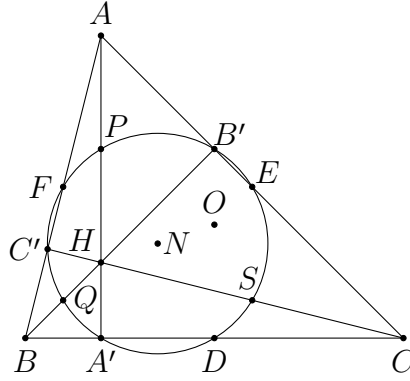
$$\begin{aligned} &2p^2q - 2abcp\bar{q} \\ &= p^2(a + b + c + p) - abcp\overline{(a + b + c + p)} \\ &= p^2(a + b + c + p) - abcp\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{p}\right) \\ &= p^3 + (a + b + c)p^2 - (ab + bc + ca)p - abc. \end{aligned}$$

This shows Q lies on ℓ , and hence ℓ bisects PH . □

Problem 2.2. (Nine-point circle) Let H and O be the orthocentre and the circumcentre of $\triangle ABC$ respectively. Let D, E, F be the midpoints of BC, CA, AB , and let A', B', C' be the feet of altitudes from A, B, C respectively. Let P, Q, S be the midpoints of AH, BH, CH respectively. Prove that $D, E, F, A', B', C', P, Q, S$ lie on the same circle. Also, prove that the centre of this circle is the midpoint N of HO , and its radius is half of the circumradius of $\triangle ABC$.

Analysis. Complex numbers can be used to show this well-known result in triangle geometry easily. All the steps involved are basic things that one should be familiar with in order to solve problems using complex numbers.

Solution.



Suppose the circumcircle is the unit circle. Then we easily find that $h = a + b + c$,

$$n = \frac{a + b + c}{2},$$

$$d = \frac{b + c}{2} \quad \text{and} \quad p = \frac{2a + b + c}{2}.$$

Also, we have

$$\begin{aligned} BC : \quad z + bc\bar{z} &= b + c, \\ AH : \quad z - bc\bar{z} &= a - bc\bar{a}. \end{aligned}$$

Solving these equations, we get

$$a' = \frac{a + b + c - bc\bar{a}}{2} = \frac{a^2 + ab + ac - bc}{2a}.$$

The circle with centre N and radius $\frac{1}{2}$ has equation $\left| z - \frac{a + b + c}{2} \right| = \frac{1}{2}$, i.e.

$$z\bar{z} - \frac{\bar{a} + \bar{b} + \bar{c}}{2}z - \frac{a + b + c}{2}\bar{z} + \frac{(a + b + c)(\bar{a} + \bar{b} + \bar{c}) - 1}{4} = 0.$$

Using the facts $\bar{a} = \frac{1}{a}$, $\bar{b} = \frac{1}{b}$ and $\bar{c} = \frac{1}{c}$, this becomes

$$z\bar{z} - \frac{ab + bc + ca}{2abc}z - \frac{a + b + c}{2}\bar{z} + \frac{(a + b)(b + c)(c + a)}{4abc} = 0. \quad (1)$$

By symmetry, it suffices to check that each of d, p, a' satisfies (1). We check these one by one as follows. For simplicity, we write the left-hand side of (1) as $f(z)$. Firstly, for D ,

$$\begin{aligned} f(d) &= \frac{b+c}{2} \cdot \frac{b+c}{2bc} - \frac{ab+bc+ca}{2abc} \cdot \frac{b+c}{2} - \frac{a+b+c}{2} \cdot \frac{b+c}{2bc} + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= \frac{a(b+c)^2}{4abc} - \frac{(ab+bc+ca)(b+c)}{4abc} - \frac{a(a+b+c)(b+c)}{4abc} + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= 0. \end{aligned}$$

Secondly, for P ,

$$\begin{aligned} f(p) &= \frac{2a+b+c}{2} \cdot \frac{2bc+ac+ab}{2abc} - \frac{ab+bc+ca}{2abc} \cdot \frac{2a+b+c}{2} - \frac{a+b+c}{2} \cdot \frac{2bc+ac+ab}{2abc} \\ &\quad + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= \frac{(2a+b+c)(2bc+ac+ab)}{4abc} - \frac{(ab+bc+ca)(2a+b+c)}{4abc} \\ &\quad - \frac{(a+b+c)(2bc+ac+ab)}{4abc} + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= 0. \end{aligned}$$

Thirdly, for A' ,

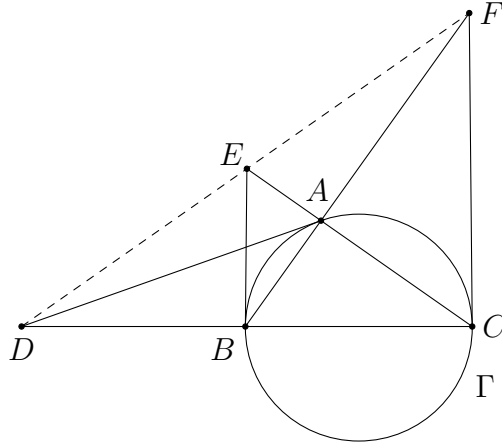
$$\begin{aligned} f(a') &= \frac{a^2+ab+ac-bc}{2a} \cdot \frac{bc+ac+ab-a^2}{2abc} - \frac{ab+bc+ca}{2abc} \cdot \frac{a^2+ab+ac-bc}{2a} \\ &\quad - \frac{a+b+c}{2} \cdot \frac{bc+ac+ab-a^2}{2abc} + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= \frac{(a^2+ab+ac-bc)(bc+ac+ab-a^2)}{4a^2bc} - \frac{(ab+bc+ca)(a^2+ab+ac-bc)}{4a^2bc} \\ &\quad - \frac{(a+b+c)(bc+ac+ab-a^2)}{4abc} + \frac{(a+b)(b+c)(c+a)}{4abc} \\ &= 0. \end{aligned}$$

The proof is complete. □

Problem 2.3. Let Γ be the circumcircle of $\triangle ABC$. The tangent at A to Γ meets the line BC at D . The tangent at B to Γ meets the line CA at E . The tangent at C to Γ meets the line AB at F . Prove that D, E, F are collinear.

Analysis. This is a special case of Pascal's theorem. The general case can also be proved using the standard setting of the unit circle. But the proof only requires some routine hard work and is not inspiring. However, that proof of Pascal's theorem has the advantage that the order of the points on the circle does not matter and it includes all the limiting cases when some of the points coincide. Here, we only present the proof of one limiting case when three pairs of points coincide.

Solution.



Suppose Γ is the unit circle. The equation of BC is

$$z + bc\bar{z} = b + c$$

and the equation of the tangent at A is

$$z + a^2 \bar{z} = 2a.$$

Solving these equations, we get

$$d = \frac{a^2(b+c) - bc(2a)}{a^2 - bc} = \frac{a^2b + a^2c - 2abc}{a^2 - bc}.$$

By symmetry, we have

$$e = \frac{b^2c + b^2a - 2abc}{b^2 - ca} \quad \text{and} \quad f = \frac{c^2a + c^2b - 2abc}{c^2 - ab}.$$

One way is to find the equation of DE and show that it is symmetric in the variables a, b, c . Alternatively, we show that $\angle DEF = 0^\circ$ or 180° by showing $\frac{f-e}{d-e}$ is real. We have

$$\begin{aligned}\frac{f-e}{d-e} &= \frac{a^2-bc}{c^2-ab} \cdot \frac{(c^2a+c^2b-2abc)(b^2-ca)-(b^2c+b^2a-2abc)(c^2-ab)}{(a^2b+a^2c-2abc)(b^2-ca)-(b^2c+b^2a-2abc)(a^2-bc)} \\ &= \frac{a^2-bc}{c^2-ab} \cdot \frac{(b-c)(a^2b^2+b^2c^2+c^2a^2-a^2bc-ab^2c-abc^2)}{(b-a)(a^2b^2+b^2c^2+c^2a^2-a^2bc-ab^2c-abc^2)} \\ &= \frac{a^2-bc}{c^2-ab} \cdot \frac{b-c}{b-a}.\end{aligned}$$

(By symmetry, $b - c$ is clearly a factor of the numerator. Then we swap a and c to get the similar factorization of the denominator.) Its conjugate is

$$\frac{\bar{a}^2 - \bar{b}\bar{c}}{\bar{c}^2 - \bar{a}\bar{b}} \cdot \frac{\bar{b} - \bar{c}}{\bar{b} - \bar{a}} = \frac{c(bc - a^2)}{a(ab - c^2)} \cdot \frac{a(c - b)}{c(a - b)} = \frac{a^2 - bc}{c^2 - ab} \cdot \frac{b - c}{b - a}.$$

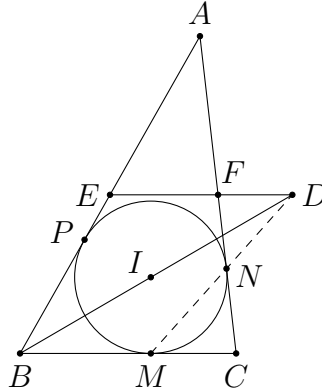
This is equal to the above expression. So $\frac{f-e}{d-e}$ is real and we are done. $\square$

The above problems are rather routine. The setup is obvious and we only need to do the algebraic works to prove the statements. But similar techniques suffice to solve problems of Olympiad level.

Problem 2.4. (CSMO 2008 Problem 6) The incircle I of $\triangle ABC$ is tangent to BC, AC at points M, N respectively. Points E, F are the midpoints of the sides AB, AC , while D is the intersection point of the lines EF and BI . Prove that M, N, D are collinear.

Analysis. We did not mention the formula for the incentre at the beginning of this section. This is because the formula does not look nice. Indeed, we can still use the same method by putting the incircle as the unit circle. Then the sides of the triangles become tangents. Everything are still easy to find.

Solution.



Suppose the incircle is the unit circle. Let AB touch the incircle at P . We know that

$$AB: \quad z + p^2 \bar{z} = 2p,$$

$$AC: \quad z + n^2 \bar{z} = 2n.$$

Solving these equations, we get

$$a = \frac{2pn}{p+n}.$$

By symmetry, we have

$$b = \frac{2pm}{p+m}, \quad c = \frac{2mn}{m+n}.$$

It follows that

$$e = \frac{1}{2} \left(\frac{2pn}{p+n} + \frac{2pm}{p+m} \right) = \frac{p(pm+pn+2mn)}{(p+n)(p+m)}, \quad f = \frac{n(np+nm+2pm)}{(n+p)(n+m)}.$$

We now find the equation of EF . Instead of following the standard procedure, we note that EF and BC are parallel by the midpoint theorem. Since the equation of BC is $z + m^2\bar{z} = 2m$, the equation of EF also has the form $z + m^2\bar{z} = \Delta$ for some expression Δ . Putting $z = e$, we find that

$$\begin{aligned}\Delta &= \frac{p(pm + pn + 2mn)}{(p+n)(p+m)} + \frac{m^2\bar{p}(\overline{pm} + \overline{pn} + 2\overline{mn})}{(\bar{p} + \bar{n})(\bar{p} + \bar{m})} \\ &= \frac{p(pm + pn + 2mn)}{(p+n)(p+m)} + \frac{m^2(n + m + 2p)}{(n+p)(m+p)} \\ &= \frac{p^2m + p^2n + 2pmn + 2pm^2 + m^3 + m^2n}{(p+m)(p+n)} \\ &= \frac{(m+p)(m+n)}{p+n}.\end{aligned}$$

Thus, we obtain

$$EF: \quad z + m^2\bar{z} = \frac{(m+p)(m+n)}{p+n}.$$

As I is the centre of the unit circle, the equation of BI is $\frac{z}{b} = \frac{\bar{z}}{\bar{b}}$, i.e.

$$BI: \quad z - pm\bar{z} = 0.$$

Solving the above two equations, we find that

$$d = \frac{p(m+p)(m+n)}{p+n} \cdot \frac{1}{p+m} = \frac{p(m+n)}{p+n}.$$

The equation of MN is $z + mn\bar{z} = m+n$. It remains to check d satisfies this equation. Indeed, we have

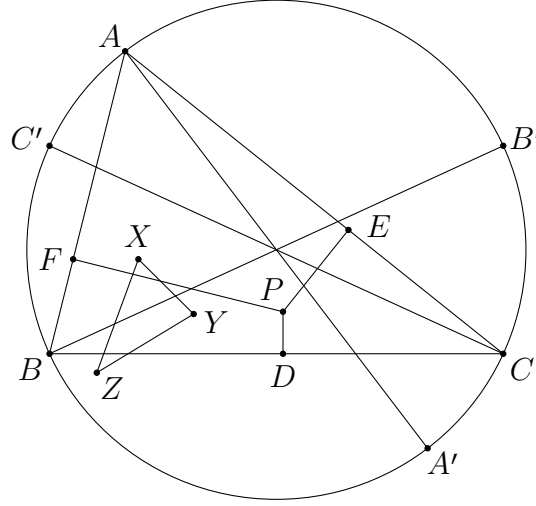
$$\frac{p(m+n)}{p+n} + mn \cdot \frac{\bar{p}(\overline{m+n})}{\bar{p} + \bar{n}} = \frac{p(m+n)}{p+n} + \frac{n(n+m)}{n+p} = m+n$$

as desired. □

Problem 2.5. (China TST 2011 Test 2 Problem 4) Let AA', BB', CC' be the three diameters of the circumcircle of acute $\triangle ABC$. P is an arbitrary point in $\triangle ABC$. The projection of the point P on BC, CA, AB are D, E, F respectively. X is the symmetric point of A' with respect to D . Y is the symmetric point of B' with respect to E . Z is the symmetric point of C' with respect to F . Prove that $\triangle XYZ \sim \triangle ABC$.

Analysis. All points can be found easily if we take the circumcircle to be the unit circle. To show that the two triangles are similar, since the figure is highly symmetric, it is more appropriate to prove it using ‘3 sides proportional’. This is not hard using complex numbers.

[Solution.](#)



Suppose (ABC) is the unit circle. Then $a' = -a$, $b' = -b$ and $c' = -c$. We have

$$BC : z + bc\bar{z} = b + c,$$

$$PD : z - bc\bar{z} = p - bc\bar{p}.$$

Solving these equations, we get

$$d = \frac{b + c + p - bc\bar{p}}{2}.$$

As D is the midpoint of $A'X$, we have

$$x = 2d - a' = a + b + c + p - bc\bar{p}.$$

By symmetry,

$$y = a + b + c + p - ca\bar{p}, \quad z = a + b + c + p - ab\bar{p}.$$

Next, we have to compute XY , etc. We find that

$$|x - y| = |c(a - b)\bar{p}| = |c||a - b||\bar{p}| = |a - b||p|$$

as $|c| = 1$. This yields $\frac{XY}{AB} = |p|$. By symmetry,

$$\frac{XY}{AB} = \frac{YZ}{BC} = \frac{ZX}{CA} = |p|.$$

This implies $\triangle ABC \sim \triangle XYZ$ as desired. $\square$

Remarks. When P is the circumcentre of $\triangle ABC$, we have $p = 0$. This gives the degenerate case where X, Y, Z coincide to the orthocentre of $\triangle ABC$.

Another way to prove that $\triangle ABC$ and $\triangle XYZ$ are directly similar is to show

$$\frac{a - b}{c - b} = \frac{x - y}{z - y}. \quad (1)$$

This is because (1) implies both $\frac{|a-b|}{|c-b|} = \frac{|x-y|}{|z-y|}$ and $\arg \frac{a-b}{c-b} = \arg \frac{x-y}{z-y}$. This means the condition ‘2 sides proportional and included angle’.

It only works for directly similarity since arguments measure directed angles. In general, $\triangle ABC \sim \triangle XYZ$ iff

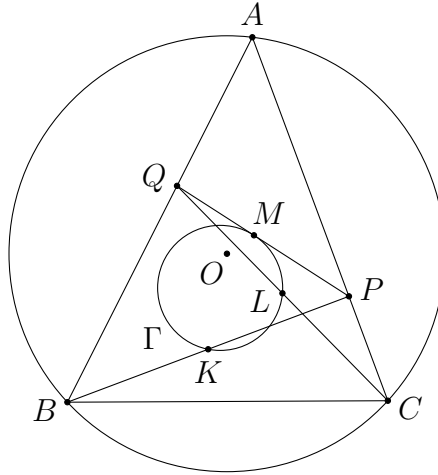
$$\frac{a-b}{c-b} = \frac{x-y}{z-y} \quad \text{or} \quad \frac{a-b}{c-b} = \frac{\bar{x}-\bar{y}}{\bar{z}-\bar{y}}.$$

This is because $\bar{x}, \bar{y}, \bar{z}$ are the reflection images of x, y, z about the real axis, hence the triangle formed by $\bar{x}, \bar{y}, \bar{z}$ is oppositely similar to the triangle formed by x, y, z .

Problem 2.6. (IMO Shortlist 2009 G2/Problem 2) Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K, L and M be the midpoints of the segments BP, CQ and PQ , respectively, and let Γ be the circle passing through K, L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

Analysis. We can either set (ABC) or Γ as the unit circle. In both ways, all points can be found easily. For the tangent line, we can formulate the condition in terms of angles. The following solution shows the application of arguments to handle angles.

Solution.



Set (ABC) as the unit circle. Let $p = sa + (1-s)c$ and $q = ta + (1-t)b$ where $s, t \in \mathbb{R}$. We easily find

$$k = \frac{sa + b + (1-s)c}{2}, \quad \ell = \frac{ta + (1-t)b + c}{2}, \quad m = \frac{(s+t)a + (1-t)b + (1-s)c}{2}.$$

Now,

$$\begin{aligned}
& PQ \text{ is tangent to } \Gamma \\
\Rightarrow & \angle KMQ = \angle KLM \\
\Rightarrow & \arg \frac{k-m}{q-m} = \arg \frac{k-\ell}{m-\ell} \\
\Rightarrow & \frac{k-m}{q-m} \div \frac{k-\ell}{m-\ell} \in \mathbb{R} \\
\Rightarrow & \frac{\frac{sa+b+(1-s)c}{2} - \frac{(s+t)a+(1-t)b+(1-s)c}{2}}{ta+(1-t)b - \frac{(s+t)a+(1-t)b+(1-s)c}{2}} \\
& \times \frac{\frac{(s+t)a+(1-t)b+(1-s)c}{2} - \frac{ta+(1-t)b+c}{2}}{\frac{sa+b+(1-s)c}{2} - \frac{ta+(1-t)b+c}{2}} \in \mathbb{R} \\
\Rightarrow & \frac{-ta+tb}{(t-s)a+(1-t)b-(1-s)c} \times \frac{sa-sc}{(s-t)a+tb-sc} \in \mathbb{R}.
\end{aligned}$$

Recall that $z \in \mathbb{R}$ iff $z = \bar{z}$. Therefore, this implies

$$\begin{aligned}
& \frac{\frac{-ta+tb}{(t-s)a+(1-t)b-(1-s)c} \times \frac{sa-sc}{(s-t)a+tb-sc}}{= \frac{\frac{-t\bar{a}+\bar{t}\bar{b}}{(t-s)\bar{a}+(1-t)\bar{b}-(1-s)\bar{c}} \times \frac{s\bar{a}-s\bar{c}}{(s-t)\bar{a}+\bar{t}\bar{b}-s\bar{c}}}. \quad (1)
\end{aligned}$$

The right-hand side is equal to

$$\frac{\frac{tc(-b+a)}{(t-s)bc+(1-t)ca-(1-s)ab} \times \frac{sb(c-a)}{(s-t)bc+tca-sab}}{$$

Thus, (1) becomes

$$\begin{aligned}
& ((t-s)bc+(1-t)ca-(1-s)ab)((s-t)bc+tca-sab) \\
& = bc((t-s)a+(1-t)b-(1-s)c)((s-t)a+tb-sc).
\end{aligned}$$

Expanding everything, we obtain the relation

$$\begin{aligned}
& s(1-s)a^2b^2 + (s^2+t^2-s-t)a^2bc + t(1-t)a^2c^2 + 2(s-t)(s+t-1)ab^2c \\
& + 2(s-t)(1-s-t)abc^2 - t(1-t)b^3c - (s^2+t^2-s-t)b^2c^2 - s(1-s)bc^3 = 0. \quad (2)
\end{aligned}$$

The left-hand side has the factor $b-c$. If this does not seem to be obvious, we may

try to find out what we need first. Indeed,

$$\begin{aligned}
OP &= OQ \\
\Leftrightarrow |sa + (1-s)c| &= |ta + (1-t)b| \\
\Leftrightarrow (sa + (1-s)c)(s\bar{a} + (1-s)\bar{c}) &= (ta + (1-t)b)(t\bar{a} + (1-t)\bar{b}) \\
\Leftrightarrow (sa + (1-s)c)\left(s\frac{1}{a} + (1-s)\frac{1}{c}\right) &= (ta + (1-t)b)\left(t\frac{1}{a} + (1-t)\frac{1}{b}\right) \\
\Leftrightarrow b(sa + (1-s)c)(sc + (1-s)a) &= c(ta + (1-t)b)(tb + (1-t)a).
\end{aligned}$$

After expansion, this is equivalent to

$$s(1-s)a^2b - t(1-t)a^2c + 2(s^2 - t^2 - s + t)abc - t(1-t)b^2c + s(1-s)bc^2 = 0. \quad (3)$$

Since this should be true under the condition (2), it should probably be a factor of that. Thus, it is now easy to observe that (2) can be rewritten as

$$(b-c)(s(1-s)a^2b - t(1-t)a^2c + 2(s^2 - t^2 - s + t)abc - t(1-t)b^2c + s(1-s)bc^2) = 0.$$

As $b \neq c$, this implies our goal (3). $\square$

Problem 2.7. (IMO Shortlist 2010 G4/Problem 2) Let I be the incentre of triangle ABC and let Γ be its circumcircle. Let the line AI intersect Γ again at D . Let E be a point on the arc $\widehat{BDC}$ and F a point on the side BC such that

$$\angle BAF = \angle CAE < \frac{1}{2} \angle BAC.$$

Finally, let G be the midpoint of the segment IF . Prove that the lines DG and EI intersect on Γ .

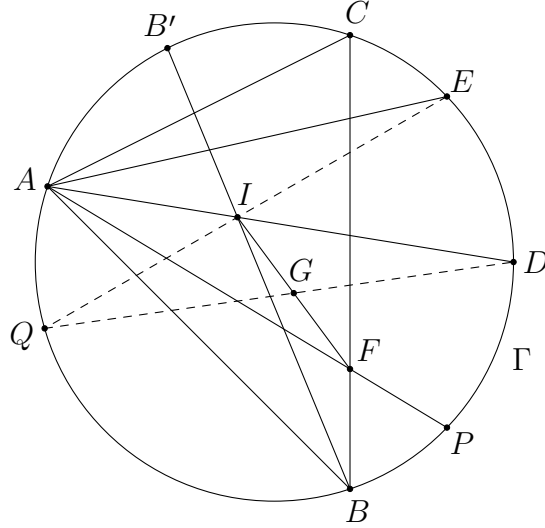
Analysis. There are 2 difficulties using the unit circle setting. Firstly, we need to handle the equal angles condition. Secondly, we need to find the complex number corresponding to the incentre I .

Note that D is the midpoint of $\widehat{BC}$. If we extend AF to meet the circumcircle at P , then D is also the midpoint of $\widehat{PE}$ since AP and AE are symmetric in the line AD . This elementary geometric fact helps us set up the figure appropriately, so that f can be found easily. Indeed, we can simply take $d = 1$ so that $b = \bar{c}$ and $p = \bar{e}$.

For the incentre, we make use of the following observation. Let X, Y, Z be points on the unit circle with Z being the midpoint of $\widehat{XY}$. Then $z^2 = xy$. To see this, we write $x = e^{i\alpha}$ and $y = e^{i\beta}$, so that $z = e^{i\frac{\alpha+\beta}{2}}$ or $z = e^{i(\frac{\alpha+\beta}{2} + \pi)}$ in view of the meaning of the polar angles. The relation $z^2 = xy$ becomes obvious under this exponential form. However, there are two such points Z satisfying the same equation, since there are two midpoints of an arc. It is not easy to distinguish the two points (note that we cannot write something like $z = \pm\sqrt{xy}$ in complex numbers), but usually this does not matter.

Now, under this observation, we can find the incentre since it is the intersection of some chords of the circumcircle, where some endpoints of the chords are midpoints of arcs.

Solution.



In this solution, we use j to denote the complex number representing I , so as to distinguish with the imaginary number i .

Let AF meet the circumcircle again at P . Let B' be the midpoint of $\widehat{AC}$ not containing B .

Suppose (ABC) is the unit circle. Without loss of generality(WLOG) we may further assume $d = 1$. Note that D is the midpoint of both $\widehat{BC}$ and $\widehat{PE}$. Therefore, we have $b = \bar{c} = \frac{1}{c}$ and $p = \bar{e} = \frac{1}{e}$.

The equation of BC is $z + bc\bar{z} = b + c$, which is

$$z + \bar{z} = c + \frac{1}{c}. \quad (1)$$

(Indeed, this is the collection of all z with the same real parts as c .) The equation of AP is $z + ap\bar{z} = a + p$, which is

$$z + \frac{a}{e}\bar{z} = a + \frac{1}{e}. \quad (2)$$

Solving (1) and (2), we find that

$$f = \frac{\frac{a}{e} \left(c + \frac{1}{c} \right) - a - \frac{1}{e}}{\frac{a}{e} - 1} = \frac{ac^2 + a - ace - c}{c(a - e)}.$$

To find j , we first note that I lies on AD , whose equation is

$$z + a\bar{z} = a + 1 \quad (3)$$

since $d = 1$. Also, I lies on BB' , whose equation is $z + bb'\bar{z} = b + b'$, i.e.

$$z + \frac{b'}{c}\bar{z} = \frac{1}{c} + b'. \quad (4)$$

Here, b' satisfies

$$(b')^2 = ac \quad (5)$$

as discussed.

From (3) and (4), we obtain

$$j = \frac{\frac{b'}{c}(a+1) - a\left(\frac{1}{c} + b'\right)}{\frac{b'}{c} - a} = \frac{ab' + b' - a - ab'c}{b' - ac}. \quad (6)$$

Then we have

$$g = \frac{f+j}{2} = \frac{(b'-ac)(ac^2 + a - ace - c) + c(a-e)(ab' + b' - a - ab'c)}{2c(a-e)(b'-ac)}. \quad (7)$$

Since the expression of g is a bit lengthy, it would be tedious to find the intersection point of EI and DG directly and claim that it lies on Γ . Thus, we define Q to be the second intersection of EI and Γ , and claim that G lies on the line DQ . This leaves the most tedious calculation to the last step.

Now, the equation of EI is $(\bar{j} - \bar{e})z - (j - e)\bar{z} + j\bar{e} - e\bar{j} = 0$, while that of Γ is $z\bar{z} = 1$. Combining these equations and cancelling $\bar{z}$, we can solve for q . Indeed, this gives

$$(\bar{j} - \bar{e})z^2 + (j\bar{e} - e\bar{j})z - (j - e) = 0.$$

This quadratic equation has solutions q and e (the intersection points of EI and Γ). By the product of roots, we have

$$q = \frac{-(j - e)}{e(\bar{j} - \bar{e})} = \frac{e - j}{e\bar{j} - 1}.$$

From (6), we find that

$$\bar{j} = \frac{\overline{ab'} + \overline{b'} - \bar{a} - \overline{ab'c}}{\overline{b'} - \bar{a}c} = \frac{c + ac - b'c - 1}{ac - b'}.$$

Therefore, one obtains

$$q = \frac{e - \frac{ab' + b' - a - ab'c}{b' - ac}}{e \cdot \frac{c + ac - b'c - 1}{ac - b'} - 1} = \frac{ace - b'e + ab' + b' - a - ab'c}{ce + ace - b'ce - e - ac + b'}. \quad (8)$$

Next, The equation of DQ is $z + q\bar{z} = 1 + q$. It remains to show

$$g + q\bar{g} = 1 + q \quad (9)$$

with g and q given by (7) and (8) respectively. Note that we still have the condition (5) unused. Thus, we should better write everything as a polynomial in b' . We first find that

$$\begin{aligned} g - 1 &= \frac{(b' - ac)(ac^2 + a - ace - c) + c(a - e)(ab' + b' - a - ab'c)}{2c(a - e)(b' - ac)} - 1 \\ &= \frac{1}{2c(a - e)(b' - ac)} \times \left(((ac^2 - 2ac + c)e + (-a^2c^2 + a^2c + ac^2 - ac + a - c))b' \right. \\ &\quad \left. + ((a^2c^2 - 2ac^2 + ac)e + (-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2)) \right) \end{aligned}$$

and

$$\begin{aligned} \bar{g} - 1 &= \frac{1}{2\bar{c}(\bar{a} - \bar{e})(\bar{b}' - \bar{ac})} \times \left(((\overline{ac^2} - \overline{2ac} + \bar{c})\bar{e} + (-\overline{a^2c^2} + \overline{a^2c} + \overline{ac^2} - \overline{ac} + \bar{a} - \bar{c}))\bar{b}' \right. \\ &\quad \left. + ((\overline{a^2c^2} - \overline{2ac^2} + \overline{ac})\bar{e} + (-\overline{a^2c^3} + \overline{2a^2c^2} - \overline{2a^2c} + \overline{ac^2})) \right) \\ &= \frac{1}{2c(e - a)(ac - b')} \times \left(((ac - 2ac^2 + a^2c^2) + (-c + c^2 + ac - ac^2 + ac^3 - a^2c^2)e) \right. \\ &\quad \left. + ((c - 2ac + ac^2) + (-1 + 2c - 2c^2 + ac)e)b' \right) \\ &= \frac{1}{2c(a - e)(b' - ac)} \times \left(((ac - 2c^2 + 2c - 1)e + (ac^2 - 2ac + c))b' \right. \\ &\quad \left. + ((-a^2c^2 + ac^3 - ac^2 + ac + c^2 - c)e + (a^2c^2 - 2ac^2 + ac)) \right). \end{aligned}$$

Hence, it suffices to show

$$\begin{aligned} &\left(((ac^2 - 2ac + c)e + (-a^2c^2 + a^2c + ac^2 - ac + a - c))b' \right. \\ &\quad \left. + ((a^2c^2 - 2ac^2 + ac)e + (-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2)) \right) \\ &+ \frac{ace - b'e + ab' + b' - a - ab'c}{ce + ace - b'ce - e - ac + b'} \times \left(((ac - 2c^2 + 2c - 1)e + (ac^2 - 2ac + c))b' \right. \\ &\quad \left. + ((-a^2c^2 + ac^3 - ac^2 + ac + c^2 - c)e + (a^2c^2 - 2ac^2 + ac)) \right) = 0, \end{aligned}$$

i.e.

$$\begin{aligned} &((-ce + 1)b' + (ac + c - 1)e - ac) \\ &\times \left(((ac^2 - 2ac + c)e + (-a^2c^2 + a^2c + ac^2 - ac + a - c))b' \right. \\ &\quad \left. + ((a^2c^2 - 2ac^2 + ac)e + (-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2)) \right) \\ &+ ((-e - ac + a + 1)b' + ace - a) \times \left(((ac - 2c^2 + 2c - 1)e + (ac^2 - 2ac + c))b' \right. \\ &\quad \left. + ((-a^2c^2 + ac^3 - ac^2 + ac + c^2 - c)e + (a^2c^2 - 2ac^2 + ac)) \right) = 0. \end{aligned}$$

When we expand the left-hand side, we get a quadratic expression in b' . Then we use (5) to replace $(b')^2$ by ac . This leaves a linear expression in b' . Since 1 and b' are independent, both coefficients should be identically 0. The checking is tedious but straightforward. The following gives the details.

The coefficient of b' is

$$\begin{aligned}
& (-ce + 1)((a^2c^2 - 2ac^2 + ac)e + (-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2)) \\
& + ((ac + c - 1)e - ac)((ac^2 - 2ac + c)e + (-a^2c^2 + a^2c + ac^2 - ac + a - c)) \\
& + (-e - ac + a + 1)((-a^2c^2 + ac^3 - ac^2 + ac + c^2 - c)e + (a^2c^2 - 2ac^2 + ac)) \\
& + (ace - a)((ac - 2c^2 + 2c - 1)e + (ac^2 - 2ac + c)) \\
& = ((-a^2c^3 + 2ac^3 - ac^2) + (a^2c^3 - 2a^2c^2 + ac^3 - 2ac^2 + 2ac + c^2 - c) \\
& + (a^2c^2 - ac^3 + ac^2 - ac - c^2 + c) + (a^2c^2 - 2ac^3 + 2ac^2 - ac))e^2 \\
& + ((a^2c^4 - 2a^2c^3 + 3a^2c^2 - ac^3 - 2ac^2 + ac) \\
& + (-a^3c^3 + a^3c^2 - a^2c^3 + 3a^2c^2 + ac^3 - 4ac^2 + 2ac - a - c^2 + c) \\
& + (a^3c^3 - a^3c^2 - a^2c^4 + 2a^2c^3 - 4a^2c^2 + a^2c + 3ac^2 - ac + c^2 - c) \\
& + (a^2c^3 - 2a^2c^2 - a^2c + 3ac^2 - 2ac + a))e \\
& + ((-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2) + (a^3c^3 - a^3c^2 - a^2c^3 + a^2c^2 - a^2c + ac^2) \\
& + (-a^3c^3 + a^3c^2 + 2a^2c^3 - 2a^2c^2 + a^2c - 2ac^2 + ac) + (-a^2c^2 + 2a^2c - ac)) \\
& = 0.
\end{aligned}$$

The constant term (after the replacement of $(b')^2$ by ac) is

$$\begin{aligned}
& ac(-ce + 1)((ac^2 - 2ac + c)e + (-a^2c^2 + a^2c + ac^2 - ac + a - c)) \\
& + ((ac + c - 1)e - ac)((a^2c^2 - 2ac^2 + ac)e + (-a^2c^3 + 2a^2c^2 - 2a^2c + ac^2)) \\
& + ac(-e - ac + a + 1)((ac - 2c^2 + 2c - 1)e + (ac^2 - 2ac + c)) \\
& + (ace - a)((-a^2c^2 + ac^3 - ac^2 + ac + c^2 - c)e + (a^2c^2 - 2ac^2 + ac)) \\
& = ((-a^2c^4 + 2a^2c^3 - ac^3) + (a^3c^3 - a^2c^3 - 2ac^3 + 3ac^2 - ac) \\
& + (-a^2c^2 + 2ac^3 - 2ac^2 + ac) + (-a^3c^3 + a^2c^4 - a^2c^3 + a^2c^2 + ac^3 - ac^2))e^2 \\
& + ((a^3c^4 - a^3c^3 - a^2c^4 + 2a^2c^3 - 3a^2c^2 + ac^3 + ac^2) \\
& + (-a^3c^4 + a^3c^3 - 2a^3c^2 - a^2c^4 + 6a^2c^3 - 5a^2c^2 + 2a^2c + ac^3 - ac^2) \\
& + (-a^3c^3 + a^3c^2 + 2a^2c^4 - 5a^2c^3 + 6a^2c^2 - a^2c - 2ac^3 + ac^2 - ac) \\
& + (a^3c^3 + a^3c^2 - 3a^2c^3 + 2a^2c^2 - a^2c - ac^2 + ac))e \\
& + ((-a^3c^3 + a^3c^2 + a^2c^3 - a^2c^2 + a^2c - ac^2) + (a^3c^4 - 2a^3c^3 + 2a^3c^2 - a^2c^3) \\
& + (-a^3c^4 + 3a^3c^3 - 2a^3c^2 - a^2c^2 + ac^2) + (-a^3c^2 + 2a^2c^2 - a^2c)) \\
& = 0.
\end{aligned}$$

This completes the proof. □

Remarks. As mentioned, there is another point B'' satisfying $(b'')^2 = ac$ other than B' . We note that the same proof still applies by using that point B'' instead of B' . In that case, the incentre I is replaced by the A -excentre of $\triangle ABC$. This happens quite often since the incentre and excentres have similar properties.

3 Basic Transformations

In pure geometry, we always apply some transformations to the figure to obtain useful results. The same things can be applied to points in the complex plane. This sometimes helps us construct the figure in a more convenient way, so that the points can be found more easily.

The transformation that we will discuss are the elementary ones, namely, translation, rotation, reflection, homothety and spiral similarity. We are supposed to be familiar with their definitions and properties. Here, we shall see how the corresponding complex numbers are changed under these transformations.

Proposition 3.1. Let AB be a directed segment. Consider the translation that moves every point P to a point P' such that PP' is the same as AB . Then

$$p' = p + (b - a).$$

Proof. We need $|p' - p| = |b - a|$ (same length) and $\arg(p' - p) = \arg(b - a)$ (same direction). These show $p' - p$ and $b - a$ have the same modulus and argument. Thus, they are equal and we have $p' - p = b - a$. $\square$

By viewing $b - a$ as a whole, this means translation is the same as adding a fixed complex number.

Proposition 3.2. Let A be a point and let θ be a real number. Consider the rotation with centre A and angle θ (anticlockwise, in radian) mapping P to P' . Then

$$p' = (p - a)\text{cis } \theta + a.$$

Proof. We need $|p' - a| = |p - a|$ (same length) and $\arg \frac{p' - a}{p - a} = \theta$ ($\angle PAP' = \theta$). The latter equation means $\frac{p' - a}{p - a} = r\text{cis } \theta$ for some $r \in \mathbb{R}^+$ since we use directed angles modulo 2π . By checking moduli, we have

$$1 = \left| \frac{p' - a}{p - a} \right| = |r\text{cis } \theta| = r\sqrt{\cos^2 \theta + \sin^2 \theta} = r.$$

This shows $p' - a = (p - a)\text{cis } \theta$. $\square$

In particular, when $a = 0$, rotation with centre 0 is simply the multiplication by $\text{cis } \theta$.

Proposition 3.3. Let AB be a straight line. Consider the reflection in the line AB mapping P to P' . Then

$$p' = \frac{a - b}{\bar{a} - \bar{b}} \bar{p} + \frac{\bar{a}b - a\bar{b}}{\bar{a} - \bar{b}}.$$

Proof. Since AB is the perpendicular bisector of PP' , we need $|p' - a| = |p - a|$ and $|p' - b| = |p - b|$. From the first equation, we have

$$(p' - a)(\overline{p'} - \overline{a}) = (p - a)(\overline{p} - \overline{a}).$$

This is the same as

$$p'\overline{p'} - \overline{a}p' - a\overline{p'} = p\overline{p} - \overline{a}p - a\overline{p},$$

which can be rewritten as

$$\overline{p'} = \frac{\overline{a}p' + p\overline{p} - \overline{a}p - a\overline{p}}{p' - a}.$$

By symmetry, this is equal to $\frac{\overline{b}p' + p\overline{p} - \overline{b}p - b\overline{p}}{p' - b}$, which yields

$$(\overline{a}p' + p\overline{p} - \overline{a}p - a\overline{p})(p' - b) = (\overline{b}p' + p\overline{p} - \overline{b}p - b\overline{p})(p' - a).$$

After expansion, we get

$$(\overline{a} - \overline{b})(p')^2 + ((\overline{b} - \overline{a})p + (b - a)\overline{p} + (a\overline{b} - \overline{a}b))p' + (a - b)p\overline{p} + (\overline{a}b - a\overline{b})p = 0.$$

Note that p also satisfies the same equation. Thus, there is a factor $p' - p$. This implies

$$(p' - p)((\overline{a} - \overline{b})p' - (a - b)\overline{p} - (\overline{a}b - a\overline{b})) = 0,$$

and hence

$$p' = \frac{a - b}{\overline{a} - \overline{b}}\overline{p} + \frac{\overline{a}b - a\overline{b}}{\overline{a} - \overline{b}}$$

as desired. $\square$

Note that the formula also holds when P lies on AB , although the proof does not imply this (as $p' = p$ in that case).

Proposition 3.4. Let A be a point and let α be a nonzero scalar. Consider the homothety with centre A and ratio α mapping P to P' . Then

$$p' = \alpha(p - a) + a.$$

Proof. Note that $p - a$ represents the directed segment AP , while $p' - a$ represents AP' . In view of the action of the homothety, we see that $AP' = \alpha AP$. This gives $p' - a = \alpha(p - a)$. $\square$

Proposition 3.5. Let A be a point, let θ be a real number and let α be a positive real number. Consider the spiral similarity with centre A , angle θ (anticlockwise, in radian) and ratio α mapping P to P' . Then

$$p' = \alpha(p - a)\text{cis } \theta + a.$$

Proof. The effect of the spiral similarity is the same as carrying out a rotation and then a homothety. By [proposition 3.2](#), the image after the rotation is

$$(p - a)\text{cis } \theta + a.$$

By [proposition 3.4](#), we get

$$p' = \alpha((p - a)\text{cis } \theta + a - a) + a = \alpha(p - a)\text{cis } \theta + a.$$

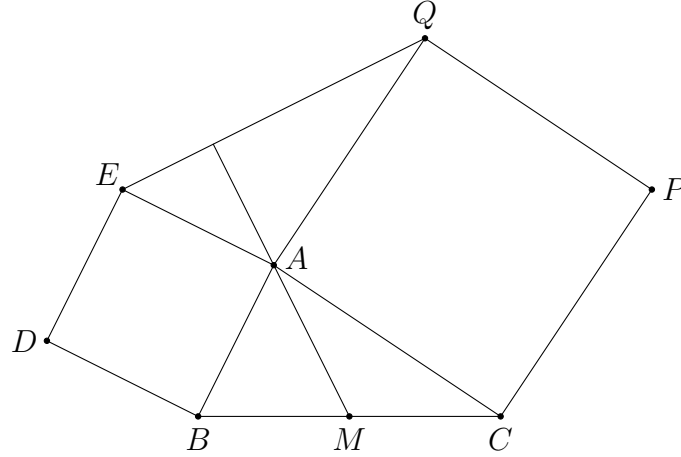
□

We shall use these facts to solve some geometric problems.

Problem 3.1. Given $\triangle ABC$, construct squares $ABDE$ and $ACPQ$ which lie outside the triangle. Let M be the midpoint of BC . Prove that $MA \perp EQ$.

Analysis. The vertices of the squares can be obtained by rotating the vertices of $\triangle ABC$ by 90° . This is one usage of the rotation formula.

Solution.



The point Q can be obtained by rotating C with centre A and angle $\frac{\pi}{2}$. So we have

$$q = (c - a)\text{cis } \frac{\pi}{2} + a = (c - a)i + a$$

by [proposition 3.2](#). Similarly, E can be obtained by rotating B with centre A and angle $-\frac{\pi}{2}$. So we have

$$e = (b - a)\text{cis } \left(-\frac{\pi}{2}\right) + a = (a - b)i + a.$$

Next, we easily get $m = \frac{b+c}{2}$. To show that $AM \perp EQ$, it suffices to show $\frac{a-m}{e-q} \in i\mathbb{R}$.

Now,

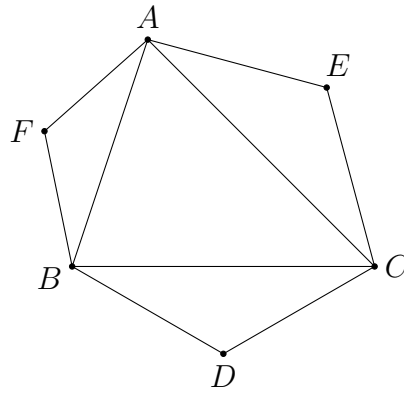
$$\frac{a-m}{e-q} = \frac{a - \frac{b+c}{2}}{(a-b)i + a - (c-a)i - a} = \frac{2a-b-c}{2(2a-b-c)i} = -\frac{1}{2}i \in i\mathbb{R}.$$

So we are done. □

Problem 3.2. (CHKMO 2010 Problem 1) Let $\triangle ABC$ be a scalene triangle. On the sides of $\triangle ABC$ three regular n -gons, external to the triangle, are drawn. Find all values of n for which the centres of the n -gons are the vertices of an equilateral triangle.

Analysis. To determine the complex numbers representing the centres of the n -gons, we can make use of rotations.

Solution.



Let D, E, F be the centres of the n -gons with sides BC, CA, AB respectively. Let $\angle CDB = \angle AEC = \angle BFA = \theta$. Note that $\theta = \frac{2\pi}{n}$. For simplicity, we write $\omega = \text{cis } \theta$.

To find d , observe that B is the rotation of C about D by angle θ . By [proposition 3.2](#), we have $b = (c - d)\omega + d$, and hence

$$d = \frac{b - c\omega}{1 - \omega}.$$

By symmetry,

$$e = \frac{c - a\omega}{1 - \omega}, \quad f = \frac{a - b\omega}{1 - \omega}.$$

Now, $\triangle DEF$ is equilateral iff F is the image of E under the rotation about D by angle $\frac{\pi}{3}$. This is equivalent to $f = (e - d)\text{cis } \frac{\pi}{3} + d$. We find that

$$\begin{aligned} f &= (e - d)\text{cis } \frac{\pi}{3} + d \\ \Leftrightarrow \frac{a - b\omega}{1 - \omega} &= \left(\frac{c - a\omega}{1 - \omega} - \frac{b - c\omega}{1 - \omega} \right) \text{cis } \frac{\pi}{3} + \frac{b - c\omega}{1 - \omega} \\ \Leftrightarrow a - b\omega &= (-a\omega - b + (1 + \omega)c) \text{cis } \frac{\pi}{3} + b - c\omega. \end{aligned}$$

This is the same as

$$\left(1 + \omega \operatorname{cis} \frac{\pi}{3}\right)a + \left(\operatorname{cis} \frac{\pi}{3} - \omega - 1\right)b + \left(\omega - (1 + \omega)\operatorname{cis} \frac{\pi}{3}\right)c = 0.$$

In order that this holds for any a, b, c (for which $|a - b|, |b - c|, |c - a|$ are distinct), we need

$$1 + \omega \operatorname{cis} \frac{\pi}{3} = \operatorname{cis} \frac{\pi}{3} - \omega - 1 = \omega - (1 + \omega)\operatorname{cis} \frac{\pi}{3} = 0. \quad (1)$$

From $1 + \omega \operatorname{cis} \frac{\pi}{3} = 0$, we can only have

$$\operatorname{cis} \frac{2\pi}{n} = \omega = -\frac{1}{\operatorname{cis} \frac{\pi}{3}} = \operatorname{cis} \frac{2\pi}{3}.$$

(For the last equality, note that $\operatorname{cis} \frac{\pi}{3} \operatorname{cis} \frac{2\pi}{3} = \operatorname{cis} \pi = -1$.) This implies

$$\frac{2\pi}{n} \equiv \frac{2\pi}{3} \pmod{2\pi}$$

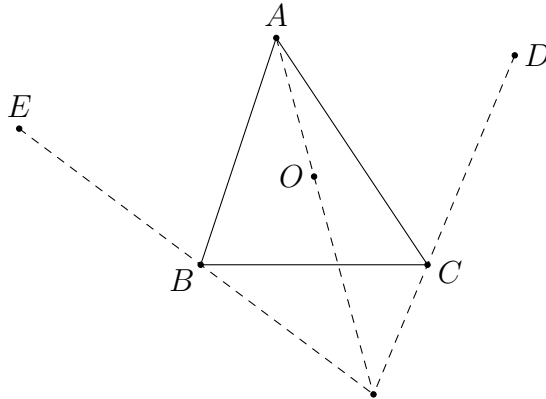
so that $n = 3$.

For $n = 3$, we have $\omega = \operatorname{cis} \frac{2\pi}{3}$, so that (1) holds by simple checking (indeed, this is a solution by the theorem of Napoleon triangle). $\square$

Problem 3.3. Let O be the circumcentre of $\triangle ABC$. Let D be the reflection of B in the line AC , and let E be the reflection of C in the line AB . Prove that BE, CD, AO are concurrent.

Analysis. This time we use the reflection formula.

Solution.



Suppose the circumcircle is the unit circle. By [proposition 3.3](#), we have

$$d = \frac{a - c}{\bar{a} - \bar{c}} \bar{b} + \frac{\bar{a}c - a\bar{c}}{\bar{a} - \bar{c}} = -\frac{ac}{b} + a + c.$$

The equation of CD is $\frac{z-c}{d-c} = \frac{\bar{z}-\bar{c}}{\bar{d}-\bar{c}}$. Now,

$$\begin{aligned} \frac{z-c}{d-c} &= \frac{\bar{z}-\bar{c}}{\bar{d}-\bar{c}} \\ \Leftrightarrow \frac{z-c}{-\frac{ac}{b}+a} &= \frac{\bar{z}-\frac{1}{c}}{-\frac{b}{ac}+\frac{1}{a}} \\ \Leftrightarrow bz+a^2c\bar{z} &= a^2+bc. \end{aligned} \quad (1)$$

The equation of AO is $\bar{a}z - a\bar{z} = 0$, i.e.

$$AO: \quad z - a^2\bar{z} = 0. \quad (2)$$

Solving (1) and (2), we find that the intersection point of CD and AO is represented by

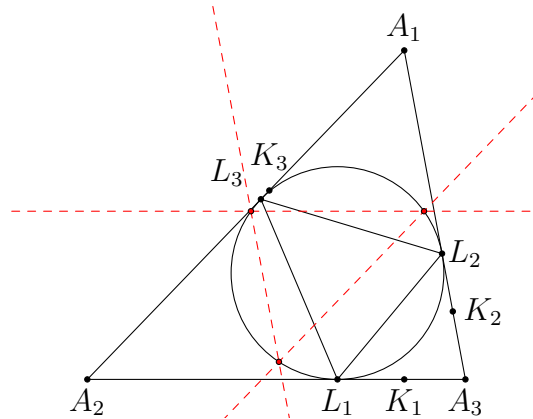
$$z = \frac{a^2+bc}{b+c}.$$

Since this is symmetric in b and c , this is also the intersection point of BE and AO . So the three lines are concurrent. $\square$

Problem 3.4. (IMO Shortlist 2000 G8/Problem 6) $A_1A_2A_3$ is an acute-angled triangle. The foot of the altitude from A_i is K_i and the incircle touches the side opposite A_i at L_i . The line K_1K_2 is reflected in the line L_1L_2 . Similarly, the line K_2K_3 is reflected in L_2L_3 and K_3K_1 is reflected in L_3L_1 . Show that the three new lines form a triangle with vertices on the incircle.

Analysis. Once again, we set the incircle as the unit circle and start from $\triangle L_1L_2L_3$. The steps are all routine, and definitely we should make use of the symmetry of the figure.

Solution.



Suppose the incircle is the unit circle. The equations of the tangents at L_1 and L_2 are $z + \ell_1^2 \bar{z} = 2\ell_1$ and $z + \ell_2^2 \bar{z} = 2\ell_2$ respectively. Solving these, we get

$$a_3 = \frac{2\ell_1\ell_2^2 - 2\ell_1^2\ell_2}{\ell_2^2 - \ell_1^2} = \frac{2\ell_1\ell_2}{\ell_1 + \ell_2}.$$

The equation of L_3K_3 is $z + \ell_3^2 \bar{z} = 2\ell_3$. Also, the equation of A_3K_3 is

$$z - \ell_3^2 \bar{z} = a_3 - \ell_3^2 \bar{a}_3 = \frac{2\ell_1\ell_2}{\ell_1 + \ell_2} - \ell_3^2 \cdot \frac{2\overline{\ell_1\ell_2}}{\overline{\ell_1 + \ell_2}} = \frac{2(\ell_1\ell_2 - \ell_3^2)}{\ell_1 + \ell_2}.$$

Solving these equations, we get

$$k_3 = \frac{\ell_3(\ell_1 + \ell_2) + \ell_1\ell_2 - \ell_3^2}{\ell_1 + \ell_2} = \frac{\ell_1\ell_2 + \ell_2\ell_3 + \ell_3\ell_1 - \ell_3^2}{\ell_1 + \ell_2}.$$

By symmetry, $k_1 = \frac{\ell_1\ell_2 + \ell_2\ell_3 + \ell_3\ell_1 - \ell_1^2}{\ell_2 + \ell_3}$.

For any point P , its reflection in the line L_3L_1 is given by $\frac{\ell_3 - \ell_1}{\ell_3 - \ell_1} \bar{p} + \frac{\overline{\ell_3\ell_1} - \ell_3\overline{\ell_1}}{\overline{\ell_3 - \ell_1}}$ using [proposition 3.3](#). This can be simplified to $-\ell_1\ell_3\bar{p} + \ell_1 + \ell_3$. Hence, the reflection K'_1 of K_1 in L_3L_1 satisfies

$$\begin{aligned} k'_1 &= -\ell_1\ell_3 \cdot \frac{\overline{\ell_1\ell_2} + \overline{\ell_2\ell_3} + \overline{\ell_3\ell_1} - \overline{\ell_1^2}}{\overline{\ell_2} + \overline{\ell_3}} + \ell_1 + \ell_3 \\ &= -\ell_1\ell_3 \cdot \frac{\ell_1\ell_3 + \ell_1^2 + \ell_1\ell_2 - \ell_2\ell_3}{\ell_1^2(\ell_3 + \ell_2)} + \ell_1 + \ell_3 \\ &= \frac{-\ell_3(\ell_1\ell_3 + \ell_1^2 + \ell_1\ell_2 - \ell_2\ell_3)}{\ell_1(\ell_2 + \ell_3)} + \frac{\ell_1(\ell_2 + \ell_3)(\ell_1 + \ell_3)}{\ell_1(\ell_2 + \ell_3)} \\ &= \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)}. \end{aligned}$$

Similarly, the reflection K'_3 of K_3 in L_3L_1 is given by $k'_3 = \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_3(\ell_1 + \ell_2)}$. We consider the line joining K'_1 and K'_3 .

$$\begin{aligned} \frac{z - k'_1}{k'_3 - k'_1} &= \frac{\bar{z} - \overline{k'_1}}{\overline{k'_3} - \overline{k'_1}} \\ \Leftrightarrow \frac{z - \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)}}{\frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_3(\ell_1 + \ell_2)} - \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)}} &= \frac{\bar{z} - \frac{\overline{\ell_2(\ell_1^2 + \ell_3^2)}}{\overline{\ell_1(\ell_2 + \ell_3)}}}{\frac{\overline{\ell_2(\ell_1^2 + \ell_3^2)}}{\overline{\ell_3(\ell_1 + \ell_2)}} - \frac{\overline{\ell_2(\ell_1^2 + \ell_3^2)}}{\overline{\ell_1(\ell_2 + \ell_3)}}} \end{aligned}$$

$$\begin{aligned}
& z - \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)} = \bar{z} - \frac{\ell_3^2 + \ell_1^2}{\ell_1\ell_3(\ell_3 + \ell_2)} \\
\Leftrightarrow & \frac{\frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_3(\ell_1 + \ell_2)} - \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)}}{\frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_3(\ell_1 + \ell_2)} - \frac{\ell_2(\ell_1^2 + \ell_3^2)}{\ell_1(\ell_2 + \ell_3)}} = \frac{\frac{\ell_3^2 + \ell_1^2}{\ell_1\ell_3(\ell_2 + \ell_1)} - \frac{\ell_3^2 + \ell_1^2}{\ell_1\ell_3(\ell_3 + \ell_2)}}{\frac{\ell_3^2 + \ell_1^2}{\ell_1\ell_3(\ell_2 + \ell_1)} - \frac{\ell_3^2 + \ell_1^2}{\ell_1\ell_3(\ell_3 + \ell_2)}} \\
\Leftrightarrow & \frac{\ell_3(\ell_1 + \ell_2)(\ell_1(\ell_2 + \ell_3)z - \ell_2(\ell_1^2 + \ell_3^2))}{\ell_2(\ell_1^2 + \ell_3^2)(\ell_1(\ell_2 + \ell_3) - \ell_3(\ell_1 + \ell_2))} = \frac{(\ell_1 + \ell_2)(\ell_1\ell_3(\ell_2 + \ell_3)\bar{z} - (\ell_1^2 + \ell_3^2))}{(\ell_1^2 + \ell_3^2)((\ell_2 + \ell_3) - (\ell_1 + \ell_2))} \\
\Leftrightarrow & \frac{\ell_3(\ell_1 + \ell_2)(\ell_1(\ell_2 + \ell_3)z - \ell_2(\ell_1^2 + \ell_3^2))}{\ell_2^2(\ell_1^2 + \ell_3^2)(\ell_1 - \ell_3)} = \frac{(\ell_1 + \ell_2)(\ell_1\ell_3(\ell_2 + \ell_3)\bar{z} - (\ell_1^2 + \ell_3^2))}{(\ell_1^2 + \ell_3^2)(\ell_3 - \ell_1)} \\
\Leftrightarrow & \ell_3(\ell_1(\ell_2 + \ell_3)z - \ell_2(\ell_1^2 + \ell_3^2)) = -\ell_2^2(\ell_1\ell_3(\ell_2 + \ell_3)\bar{z} - (\ell_1^2 + \ell_3^2)) \\
\Leftrightarrow & \ell_1\ell_3(\ell_2 + \ell_3)z + \ell_1\ell_2^2\ell_3(\ell_2 + \ell_3)\bar{z} = \ell_2(\ell_2 + \ell_3)(\ell_1^2 + \ell_3^2) \\
\Leftrightarrow & \ell_1\ell_3z + \ell_1\ell_2^2\ell_3\bar{z} = \ell_2(\ell_1^2 + \ell_3^2)
\end{aligned}$$

By symmetry, the line obtained by reflecting K_1K_2 in L_1L_2 has equation

$$\ell_1\ell_2z + \ell_1\ell_2\ell_3^2\bar{z} = \ell_3(\ell_1^2 + \ell_2^2).$$

Solving the above two equations, the complex number corresponding to their intersection is

$$z = \frac{\ell_2\ell_3(\ell_1^2 + \ell_3^2) - \ell_2\ell_3(\ell_1^2 + \ell_2^2)}{\ell_1(\ell_3^2 - \ell_2^2)} = \frac{\ell_2\ell_3}{\ell_1}.$$

Since $\left|\frac{\ell_2\ell_3}{\ell_1}\right| = 1$, this intersection point does lie on the incircle. This completes the proof by symmetry. $\square$

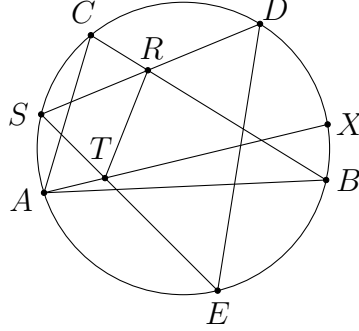
Problem 3.5. (CMO 2011 Problem 2) In the circumcircle of an acute triangle ABC , point D is the midpoint of the arc $\widehat{BC}$, X lies on the arc $\widehat{BD}$, E is the midpoint of the arc $\widehat{AX}$, while S lies on the arc $\widehat{AC}$. The lines SD and BC intersect at R . The lines SE and AX intersect at T . Prove that if $RT \parallel DE$, then the incentre of triangle ABC lies on the line RT .

Analysis. We always make use of the symmetry of the figure to simplify our calculation. Although this problem is not set in a symmetric way, one should find pairs of corresponding points in the figure which behave the same. Indeed, we note that $(A, C), (X, B), (E, D), (T, R)$ are those pairs, while point S seems to be a singleton.

With this observation, we should reconstruct the points starting from S so that the symmetry can be used. One should bear in mind that it is not necessary to start from $\triangle ABC$ every time.

To determine the incentre, this time we use a well-known property of the incentre I and the circumcircle, i.e. I lies on AD and $DI = DB = DC$.

Solution.



Suppose (ABC) is the unit circle. Since $RT \parallel DE$, there exists a homothety with centre S and a certain ratio $\alpha \in \mathbb{R}$ mapping D, E to R, T respectively. Thus,

$$r = \alpha(d - s) + s, \quad t = \alpha(e - s) + s$$

by [proposition 3.4](#).

Since D is the midpoint of $\widehat{BC}$, BC is parallel to the tangent at D , whose equation is $z + d^2\bar{z} = 2d$. Thus, BC has the form $z + d^2\bar{z} = \Delta_1$. As R lies on BC , we have

$$\begin{aligned} \Delta_1 &= (\alpha(d - s) + s) + d^2(\alpha(\bar{d} - \bar{s}) + \bar{s}) \\ &= (\alpha(d - s) + s) + d^2\left(\alpha\left(\frac{1}{d} - \frac{1}{s}\right) + \frac{1}{s}\right) \\ &= -\frac{(s - d)^2}{s}\alpha + \frac{s^2 + d^2}{s}. \end{aligned}$$

Hence, the equation of BC is

$$z + d^2\bar{z} = -\frac{(s - d)^2}{s}\alpha + \frac{s^2 + d^2}{s}.$$

To solve for b, c , we combine the equation of BC and that of (ABC) , i.e. $z\bar{z} = 1$. After rearranging the terms, we need to solve

$$z^2 + \frac{(s - d)^2\alpha - (s^2 + d^2)}{s}z + d^2 = 0. \quad (1)$$

In other words, b, c are the roots of the quadratic equation (1). There is no need to solve the equation explicitly, since we only need to make use of (1) later. By symmetry, a, x are the roots of

$$z^2 + \frac{(s - e)^2\alpha - (s^2 + e^2)}{s}z + e^2 = 0. \quad (2)$$

Let J be the intersection of RT and AD . Our goal is to show that $J = I$, the incentre

of $\triangle ABC$. Firstly, we simplify the equation of RT .

$$\begin{aligned}
& \frac{z-r}{t-r} = \frac{\bar{z}-\bar{r}}{\bar{t}-\bar{r}} \\
\Leftrightarrow & \frac{z - (\alpha(d-s) + s)}{(\alpha(e-s) + s) - (\alpha(d-s) + s)} = \frac{\bar{z} - (\alpha(\bar{d}-\bar{s}) + \bar{s})}{(\alpha(\bar{e}-\bar{s}) + \bar{s}) - (\alpha(\bar{d}-\bar{s}) + \bar{s})} \\
\Leftrightarrow & \frac{z + (s-d)\alpha - s}{(e-d)\alpha} = \frac{\bar{z} + (\bar{s}-\bar{d})\alpha - \bar{s}}{(\bar{e}-\bar{d})\alpha} \\
\Leftrightarrow & \frac{z + (s-d)\alpha - s}{(e-d)\alpha} = \frac{sde\bar{z} + e(d-s)\alpha - de}{s(d-e)\alpha} \\
\Leftrightarrow & sz + s(s-d)\alpha - s^2 = -(sde\bar{z} + e(d-s)\alpha - de) \\
\Leftrightarrow & sz + sde\bar{z} = -(s-d)(s-e)\alpha + (s^2 + de)
\end{aligned}$$

The equation of AD is $z + ad\bar{z} = a + d$. Solving these, we find

$$j = \frac{-a(s-d)(s-e)\alpha + s^2a - sae - sde + ade}{s(a-e)}.$$

Now, we show that $DJ = DB$. Indeed,

$$\begin{aligned}
& DJ = DB \\
\Leftrightarrow & |d-j| = |d-b| \\
\Leftrightarrow & \left| \frac{a(s-d)(s-e)\alpha - s^2a + sad + sae - ade}{s(a-e)} \right| = |d-b| \\
\Leftrightarrow & \left| \frac{a(s-d)(s-e)\alpha - a(s-d)(s-e)}{s(a-e)} \right| = |d-b| \\
\Leftrightarrow & \left| \frac{(s-d)(s-e)}{s(a-e)} \right| |\alpha - 1| = |d-b| \\
\Leftrightarrow & \frac{(s-d)(s-e)}{s(a-e)} \cdot \frac{(\bar{s}-\bar{d})(\bar{s}-\bar{e})}{\bar{s}(\bar{a}-\bar{e})} (\alpha-1)^2 = (d-b)(\bar{d}-\bar{b}) \\
\Leftrightarrow & \frac{(s-d)(s-e)}{s(a-e)} \cdot \frac{a(d-s)(e-s)}{sd(e-a)} (\alpha-1)^2 = (d-b) \cdot \frac{b-d}{bd} \\
\Leftrightarrow & \frac{(s-d)^2(s-e)^2}{s^2} \cdot (\alpha-1)^2 \cdot \frac{a}{(a-e)^2} = \frac{(b-d)^2}{b}.
\end{aligned}$$

We now use (2) to eliminate the term a in the fraction $\frac{a}{(a-e)^2}$. Indeed,

$$\begin{aligned}
\frac{a}{(a-e)^2} &= \frac{a}{a^2 + e^2 - 2ea} = \frac{a}{\frac{-(s-e)^2\alpha + (s^2 + e^2)}{s} - a - 2ea} \\
&= \frac{s}{-(s-e)^2\alpha + (s^2 + e^2) - 2se} = \frac{s}{(s-e)^2(1-\alpha)}.
\end{aligned}$$

By symmetry, we find that

$$\frac{(b-d)^2}{b} = \frac{(s-d)^2(1-\alpha)}{s}.$$

Combining all these, $DJ = DB$ holds iff

$$\frac{(s-d)^2(s-e)^2}{s^2} \cdot (\alpha-1)^2 \cdot \frac{s}{(s-e)^2(1-\alpha)} = \frac{(s-d)^2(1-\alpha)}{s}.$$

This is clearly true.

Although there are 2 possible locations of J (the incentre and the A -excentre of $\triangle ABC$), it is clear that J lies inside $\triangle ABC$ from the configuration (a figure is given in the CMO). Hence, J is the incentre of $\triangle ABC$. $\square$

Remarks. If the order of the points on the circle is not specified, then it is possible that J is an excentre. Also, the same argument also applies to $\triangle XCB$ instead of $\triangle ACB$ since A, X have the same role.

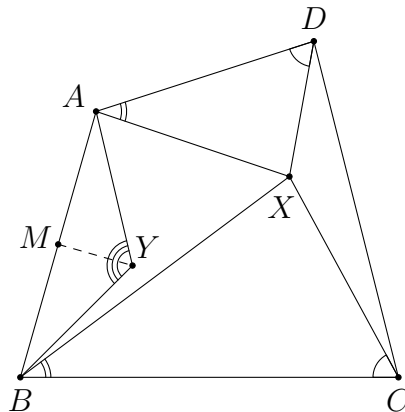
Problem 3.6. (IMO Shortlist 2000 G6) Let $ABCD$ be a convex quadrilateral. The perpendicular bisectors of its sides AB and CD meet at Y . Denote by X a point inside the quadrilateral $ABCD$ such that

$$\angle ADX = \angle BCX < 90^\circ \quad \text{and} \quad \angle DAX = \angle CBX < 90^\circ.$$

Show that $\angle AYB = 2\angle ADX$.

Analysis. It is obvious that $\triangle AXD \sim \triangle BXC$. To apply this condition on the complex plane, we can use some geometric transformations to take one triangle to the other. Since the orientation of the similarity is reversed, we need a spiral similarity followed by a reflection. The simplest reflection is to reflect in the real axis.

Solution.



WLOG assume $x = 0$. Let A', D' be the reflections of A, D in the real axis respectively. Then $a' = \bar{a}$ and $d' = \bar{d}$.

Since $\triangle XA'D'$ is directly similar to $\triangle XBC$, there exists a spiral similarity with centre X mapping A', D' to B, C respectively. Let $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{R}$ be the angle and the ratio of this spiral similarity. Define $\omega = \text{cis } \theta$. Then $b = \alpha\omega\bar{a}$ and $c = \alpha\omega\bar{d}$ by [proposition 3.5](#).

Consider the perpendicular bisector of AB .

$$\begin{aligned} & |z - a| = |z - b| \\ \Leftrightarrow & (z - a)(\bar{z} - \bar{a}) = (z - b)(\bar{z} - \bar{b}) \\ \Leftrightarrow & (z - a)(\bar{z} - \bar{a}) = (z - \alpha\omega\bar{a})\left(\bar{z} - \frac{\alpha}{\omega}a\right) \\ \Leftrightarrow & \left(\frac{\alpha}{\omega}a - \bar{a}\right)z + (\alpha\omega\bar{a} - a)\bar{z} = (\alpha^2 - 1)a\bar{a} \end{aligned}$$

By symmetry, the perpendicular bisector of CD has equation

$$\left(\frac{\alpha}{\omega}d - \bar{d}\right)z + (\alpha\omega\bar{d} - d)\bar{z} = (\alpha^2 - 1)d\bar{d}.$$

Solving these equations, we get

$$\begin{aligned} y &= \frac{(\alpha^2 - 1)(\alpha\omega\bar{a} - a)d\bar{d} - (\alpha^2 - 1)(\alpha\omega\bar{d} - d)a\bar{a}}{\left(\frac{\alpha}{\omega}d - \bar{d}\right)(\alpha\omega\bar{a} - a) - \left(\frac{\alpha}{\omega}a - \bar{a}\right)(\alpha\omega\bar{d} - d)} \\ &= \frac{(\alpha^2 - 1)(\alpha\omega\bar{a}d(d - a) + ad(\bar{a} - \bar{d}))}{\alpha^2(\bar{a}d - a\bar{d}) + (a\bar{d} - \bar{a}d)} \\ &= \frac{\alpha\omega\bar{a}d(d - a) + ad(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}}. \end{aligned}$$

Next, to show that $\angle AYB$ is twice of $\angle ADX$, we can first bisect $\angle AYB$. Denote by M the midpoint of AB . Then $m = \frac{\alpha\omega\bar{a} + a}{2}$. Note that $YA = YB$. This implies $\angle AYM = \frac{1}{2}\angle AYB$. Thus, it suffices to prove $\angle AYM = \angle ADX$. Now,

$$\begin{aligned} & \angle AYM = \angle ADX \\ \Leftrightarrow & \arg \frac{m - y}{a - y} = \arg \frac{x - d}{a - d} \\ \Leftrightarrow & \arg \frac{\frac{\alpha\omega\bar{a} + a}{2} - \frac{\alpha\omega\bar{a}d(d - a) + ad(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}}}{a - \frac{\alpha\omega\bar{a}d(d - a) + ad(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}}} = \arg \frac{-d}{a - d} \end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow \arg \frac{\alpha\omega\bar{a}(\bar{a}d + a\bar{d} - 2d\bar{d}) + a(2d\bar{d} - \bar{a}d - a\bar{d})}{2(\alpha\omega\bar{a}d(a-d) + a\bar{d}(d-a))} = \arg \frac{d}{d-a} \\
&\Leftrightarrow \arg \frac{(a - \alpha\omega\bar{a})(2d\bar{d} - \bar{a}d - a\bar{d})}{2\bar{d}(d-a)(a - \alpha\omega\bar{a})} = \arg \frac{d}{d-a} \\
&\Leftrightarrow \arg \frac{2d\bar{d} - \bar{a}d - a\bar{d}}{2\bar{d}(d-a)} = \arg \frac{d}{d-a} \\
&\Leftrightarrow \frac{2d\bar{d} - \bar{a}d - a\bar{d}}{2\bar{d}(d-a)} \div \frac{d}{d-a} \in \mathbb{R} \\
&\Leftrightarrow \frac{2d\bar{d} - \bar{a}d - a\bar{d}}{2d\bar{d}} \in \mathbb{R} \\
&\Leftrightarrow \frac{2d\bar{d} - \bar{a}d - a\bar{d}}{2d\bar{d}} = \frac{2\bar{d}d - a\bar{d} - \bar{a}d}{2\bar{d}d}.
\end{aligned}$$

This is obviously true. □

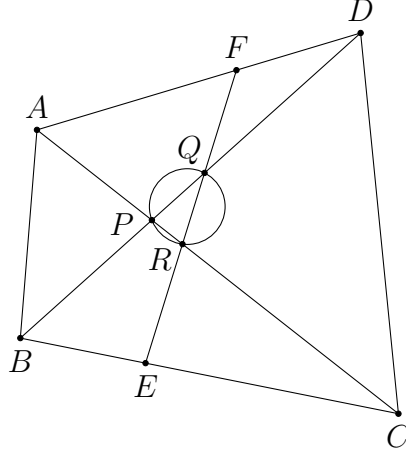
Remarks. Strictly speaking, we need $\frac{2d\bar{d} - \bar{a}d - a\bar{d}}{2d\bar{d}} \in \mathbb{R}^+$ to show that the two angles are equal. If it is a real number, then we only know that the two angles are equal modulo π . But since we know that both angles are less than π from the configuration, they must be equal.

Problem 3.7. (IMO Shortlist 2005 G4/Problem 5) Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie on the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .

Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

Analysis. This time we can use rotation to send the points A, F, D to C, E, B respectively. The centre of the rotation is the intersection of the perpendicular bisectors of AC and DB . We shall set this centre as O due to symmetry. But then the collinearity of A, F, D (and hence C, E, B) does not give a simple equation. We shall leave this condition unused until the last part of the solution.

Solution. Let $O(0)$ be the intersection of the perpendicular bisectors of AC and DB . Since $\triangle OAD \cong \triangle OCB$, there exists a rotation about O which maps A to C and D to B . In addition, since $BE = DF$, the rotation also maps F to E . Let θ be the angle of the rotation, and let $\omega = \text{cis } \theta$. Then we have $c = \omega a$, $e = \omega f$ and $b = \omega d$.



The equation of AC is $\frac{z-a}{\omega a-a} = \frac{\bar{z}-\bar{a}}{\bar{\omega}\bar{a}-\bar{a}}$. Since we have $\bar{\omega} = \frac{1}{\omega}$, this is the same as $\frac{z-a}{(\omega-1)a} = \frac{\bar{z}-\bar{a}}{\frac{1}{\omega}(1-\omega)\bar{a}}$, i.e.

$$AC: \quad \bar{a}z + \omega a\bar{z} = (\omega+1)a\bar{a}.$$

By symmetry,

$$BD: \quad \bar{d}z + \omega d\bar{z} = (\omega+1)d\bar{d}.$$

Solving these equations, we get $p = \frac{(\omega+1)ad(\bar{a}-\bar{d})}{\bar{a}d - a\bar{d}}$. Similarly, $q = \frac{(\omega+1)df(\bar{d}-\bar{f})}{\bar{d}f - d\bar{f}}$

and $r = \frac{(\omega+1)af(\bar{a}-\bar{f})}{\bar{a}f - a\bar{f}}$.

To find (PQR) , we first find the circumcentre S of $\triangle PQR$. The perpendicular bisector of PQ has equation $|z-p| = |z-q|$. We find that

$$\begin{aligned} & |z-p| = |z-q| \\ \Leftrightarrow & (z-p)(\bar{z}-\bar{p}) = (z-q)(\bar{z}-\bar{q}) \\ \Leftrightarrow & (\bar{p}-\bar{q})z + (p-q)\bar{z} = p\bar{p} - q\bar{q}. \end{aligned}$$

Also, the perpendicular bisector of PR has equation

$$(\bar{p}-\bar{r})z + (p-r)\bar{z} = p\bar{p} - r\bar{r}.$$

Solving these equations, we get

$$s = \frac{(p-r)(p\bar{p}-q\bar{q}) - (p-q)(p\bar{p}-r\bar{r})}{(p-r)(\bar{p}-\bar{q}) - (p-q)(\bar{p}-\bar{r})} = \frac{(q-r)p\bar{p} + (r-p)q\bar{q} + (p-q)r\bar{r}}{(q-r)\bar{p} + (r-p)\bar{q} + (p-q)\bar{r}}.$$

For the collinearity condition, it means $f = ta + (1 - t)d$ where $t \in \mathbb{R}$. Then we have

$$\begin{aligned} q &= \frac{(\omega + 1)d\bar{f}(\bar{d} - \bar{f})}{\bar{d}f - d\bar{f}} \\ &= \frac{(\omega + 1)d(ta + (1 - t)d)(-t\bar{a} + t\bar{d})}{ta\bar{d} + (1 - t)d\bar{d} - t\bar{a}d - (1 - t)d\bar{d}} \\ &= \frac{(\omega + 1)d(ta + (1 - t)d)(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}}. \end{aligned}$$

By interchanging a and d , and replacing t by $1 - t$, we obtain

$$r = \frac{(\omega + 1)a(ta + (1 - t)d)(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}}.$$

If we define

$$\Delta = \frac{(\omega + 1)(\bar{a} - \bar{d})}{\bar{a}d - a\bar{d}},$$

then our results are equivalent to

$$p = ad\Delta, \quad q = d(ta + (1 - t)d)\Delta, \quad r = a(ta + (1 - t)d)\Delta.$$

In the following, we give two methods to complete the proof.

Method 1. In this method, we need one important observation, i.e. to guess the desired point. Firstly, a good diagram certainly helps us to do so. Apart from that, we should choose a point which plays the same role w.r.t. the components APC and DPB due to symmetry. There are not many such points. Indeed, we shall prove that O always lies on (PQR) . Note that

$$\begin{aligned} |s| &= |s - p| \\ \Leftrightarrow \left| \frac{(q - r)p\bar{p} + (r - p)q\bar{q} + (p - q)r\bar{r}}{(q - r)\bar{p} + (r - p)\bar{q} + (p - q)\bar{r}} \right| &= \left| \frac{(q - r)p\bar{p} + (r - p)q\bar{q} + (p - q)r\bar{r}}{(q - r)\bar{p} + (r - p)\bar{q} + (p - q)\bar{r}} - p \right| \\ \Leftrightarrow \left| \frac{(q - r)p\bar{p} + (r - p)q\bar{q} + (p - q)r\bar{r}}{(q - r)\bar{p} + (r - p)\bar{q} + (p - q)\bar{r}} \right| &= \left| \frac{(r - p)(q - p)\bar{q} + (p - q)(r - p)\bar{r}}{(q - r)\bar{p} + (r - p)\bar{q} + (p - q)\bar{r}} \right| \\ \Leftrightarrow |(q - r)p\bar{p} + (r - p)q\bar{q} + (p - q)r\bar{r}| &= |(\bar{q} - \bar{r})(r - p)(q - p)| \\ \Leftrightarrow |(q - r)p\bar{p} + (r - p)q\bar{q} + (p - q)r\bar{r}| &= |(q - r)(r - p)(q - p)|. \end{aligned} \tag{1}$$

We now evaluate each term one by one. For the right-hand side, we have

$$\begin{aligned} q - r &= d(ta + (1 - t)d)\Delta - a(ta + (1 - t)d)\Delta = (d - a)(ta + (1 - t)d)\Delta, \\ r - p &= a(ta + (1 - t)d)\Delta - ad\Delta = t(a - d)a\Delta, \\ q - p &= d(ta + (1 - t)d)\Delta - ad\Delta = (1 - t)(d - a)d\Delta. \end{aligned}$$

These give

$$(q - r)(r - p)(q - p) = t(1 - t)(a - d)^3(ta + (1 - t)d)ad\Delta^3. \tag{2}$$

For the left-hand side, we have

$$\begin{aligned}(q-r)p\bar{p} &= (d-a)(ta+(1-t)d)ad\Delta^2\overline{ad\Delta}, \\ (r-p)q\bar{q} &= t(a-d)(ta+(1-t)d)(t\bar{a}+(1-t)\bar{d})ad\Delta^2\overline{d\Delta}, \\ (p-q)r\bar{r} &= (1-t)(a-d)(ta+(1-t)d)(t\bar{a}+(1-t)\bar{d})ad\Delta^2\overline{a\Delta}.\end{aligned}$$

These give

$$\begin{aligned}& (q-r)p\bar{p} + (r-p)q\bar{q} + (p-q)r\bar{r} \\ &= (a-d)(ta+(1-t)d)ad\Delta^2\overline{\Delta}\left(-\overline{ad} + t(t\bar{a}+(1-t)\bar{d})\bar{d} + (1-t)(t\bar{a}+(1-t)\bar{d})\bar{a}\right) \\ &= (a-d)(ta+(1-t)d)ad\Delta^2\overline{\Delta}t(1-t)(\bar{a}-\bar{d})^2.\end{aligned}\tag{3}$$

From (1), (2) and (3), it remains to prove

$$|t(1-t)(a-d)(\bar{a}-\bar{d})^2(ta+(1-t)d)ad\Delta^2\overline{\Delta}| = |t(1-t)(a-d)^3(ta+(1-t)d)ad\Delta^3|.$$

This is clear from the properties of moduli. Thus, O lies on (PQR) , which is a fixed point as E, F vary. Note that the condition that BC and DA are not parallel means P and O are distinct.

Method 2. In this method, we do not need to guess the desired point, but the steps are more tedious as expected. Indeed, we shall show that the locus of the centre S is a straight line ℓ . Once this is proved, we know that (PQR) always passes through the reflection of P in the line ℓ , since that point is fixed and it always has the same distance as P to S .

To find the locus of S as F varies, we should better express

$$s = \frac{(q-r)p\bar{p} + (r-p)q\bar{q} + (p-q)r\bar{r}}{(q-r)\bar{p} + (r-p)\bar{q} + (p-q)\bar{r}}$$

in terms of a, f, d . We find that

$$\begin{aligned}(q-r)\bar{p} &= (d-a)(ta+(1-t)d)\Delta\overline{ad\Delta}, \\ (r-p)\bar{q} &= t(a-d)(t\bar{a}+(1-t)\bar{d})a\Delta\overline{d\Delta}, \\ (p-q)\bar{r} &= (1-t)(a-d)(t\bar{a}+(1-t)\bar{d})d\Delta\overline{a\Delta}.\end{aligned}$$

These give

$$\begin{aligned}& (q-r)\bar{p} + (r-p)\bar{q} + (p-q)\bar{r} \\ &= (a-d)\Delta\overline{\Delta}\left(-(ta+(1-t)d)\overline{ad} + t(t\bar{a}+(1-t)\bar{d})a\bar{d} + (1-t)(t\bar{a}+(1-t)\bar{d})d\bar{a}\right) \\ &= (a-d)\Delta\overline{\Delta}\left(t(1-t)\overline{a^2d} - t(1-t)a\overline{ad} - t(1-t)d\overline{ad} + t(1-t)\overline{ad^2}\right) \\ &= t(1-t)(a-d)(\bar{a}-\bar{d})(\bar{a}d - a\bar{d})\Delta\overline{\Delta}.\end{aligned}$$

Together with (3), we get

$$s = \frac{t(1-t)(a-d)(\bar{a}-\bar{d})^2(ta+(1-t)d)ad\Delta^2\bar{\Delta}}{t(1-t)(a-d)(\bar{a}-\bar{d})(\bar{a}d-a\bar{d})\Delta\bar{\Delta}} = \frac{(\bar{a}-\bar{d})(ta+(1-t)d)ad\Delta}{\bar{a}d-a\bar{d}}.$$

Now, this shows s and t are linearly related. Since BC and DA are not parallel, the coefficient of t is nonzero. Thus, we can write $t = \alpha s + \beta$ for some $\alpha \neq 0$ and β depending only on a, d, θ (which means only on $ABCD$). As

$$\alpha s + \beta = t = \bar{t} = \overline{\alpha s + \beta},$$

we obtain $\alpha s - \overline{\alpha s} = \bar{\beta} - \beta = -2\text{Im}(\beta)i$. This shows the locus of S is a straight line. However, we also need to verify the second common point that (PQR) passes through is not P . Suppose this is not true. Then P lies on the locus of S , which means there exists $t \in \mathbb{R}$ such that

$$\frac{(\bar{a}-\bar{d})(ta+(1-t)d)ad\Delta}{\bar{a}d-a\bar{d}} = ad\Delta.$$

Note that

$$\begin{aligned} & \frac{(\bar{a}-\bar{d})(ta+(1-t)d)ad\Delta}{\bar{a}d-a\bar{d}} = ad\Delta \\ \Leftrightarrow & (\bar{a}-\bar{d})(ta+(1-t)d) = \bar{a}d-a\bar{d} \\ \Leftrightarrow & t(a-d)(\bar{a}-\bar{d}) = (d-a)\bar{d} \\ \Leftrightarrow & \frac{\bar{d}}{\bar{d}-\bar{a}} = t \in \mathbb{R} \\ \Leftrightarrow & \frac{\bar{d}}{\bar{d}-\bar{a}} = \frac{d}{d-a} \\ \Leftrightarrow & \bar{a}d = a\bar{d} \\ \Leftrightarrow & \bar{a}d \in \mathbb{R} \\ \Leftrightarrow & \frac{d}{a} \in \mathbb{R}. \end{aligned}$$

This is the same as A, O, D being collinear. By symmetry, we also need B, O, C being collinear. This cannot hold since $ABCD$ is convex. □

Remarks. Both methods are doable. Even for the same setting, there are various ways to complete the proof. Some may require stronger observations, while some may require more geometric knowledge.

Links

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Problems

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- [3.1](#): Rotation
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- [3.5](#): CMO 2011 Problem 2
- [3.6](#): IMO Shortlist 2000 G6

- 3.7: IMO 2005 Problem 5

Terminologies and Notations

- $\mathbb{C}$
- $i\mathbb{R}$
- Argand diagram
- argument $\arg z$
- complex conjugate $\bar{z}$
- complex number
- complex plane
- exponential form $re^{i\theta}$
- imaginary number
- imaginary part $\text{Im}(z)$
- modulus $|z|$
- polar form $r\text{cis } \theta$
- purely imaginary number
- real part $\text{Re}(z)$
- roots of unity
- standard form $a + bi$

References

- Mathematical Excalibur
(Available from: http://www.math.ust.hk/excalibur/v1_n3.pdf
http://www.math.ust.hk/excalibur/v9_n1.pdf
http://www.math.ust.hk/excalibur/v13_n3.pdf)

「持而盈之，不如其已；揣而銳之，不可常保。」

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